

IS COMPOUND INTEREST THE MOST POWERFUL FORCE IN THE UNIVERSE?

How do you grow an investment of \$1 000 into \$35.5 million over 58 years? Yet, this is what Warren Buffett managed to achieve for his shareholders. Nearer to home, Allan Gray managed to do even better by growing R10 000 into R317.5 million over 48 years for Allan Gray's investors. What is the compound return that Warren Buffett and Allan Gray achieved for their shareholders and how do their returns compare to investing in the market index? We will answer this question and analyse the formulae and application of time value of money principles to real world problems. We will also analyse the role of interest rates and determine how to value bonds.

LEARNING OBJECTIVES

After working through this chapter, you should be able to:

- Understand the role of time value of money in finance and understand the concept of compound interest.
- Use formulae, tables, financial calculators and spreadsheets to determine:
 - the future value of a single amount invested today;
 - the future value of an annuity;
 - the present value of a single future amount;
 - the present value of an annuity;
 - the present value of a perpetuity;
 - the present value of a growing perpetuity;

- the present value of a cash flow growing at a constant rate over a period of time;
 - the present value of uneven cash flow streams.
- Define and calculate an annual effective rate.
 - Distinguish between nominal and real interest rates.
 - Apply compounding and discounting to complex cash flow streams.
 - Apply time value of money principles to real world problems such as retirement planning and the valuation of bonds and
 - Establish the factors that determine the term structure of interest rates.

INTRODUCTION

In this chapter we will learn how to value estimated future cash flows and apply time value of money principles to such applications as determining the repayments on a loan. The principles outlined here are fundamental to all aspects of financial management and serve as essential tools in the activities of the financial manager.

Companies and individuals often have opportunities to earn returns on investments. When you invest in a savings account, you will earn a return, the interest rate, for a specified period. When companies invest in plant and machinery, the company is expecting to earn cash flows over a number of years. When you take out a student loan, you wish to know what the repayments will be. A financial manager is required to value projects with cash flows that will occur at different points in time. The timing of cash flows is an important component of corporate finance decisions. Why is a rand today worth more than a rand in a year's time? The simple answer is that you can invest the rand today to accumulate to a larger sum in a year's time. We are required to determine future values and present values of cash flows. To do this, we will develop formulae which we can apply to real world applications.

We can also use Tables which are found at the back of the book. However, increasingly we use financial calculators and Excel or other spreadsheet programs to solve time value of money problems. Whilst we do not often need the formulae to determine present values and future values, and we could focus only on the use of financial calculators, it is useful to understand the **Page 2-2** formulae, as these are fundamental to many applications. Further, in the real world, we often may need to set

up our own financial spreadsheet models and we may be required to use the formulae rather than use functions.

1 FUTURE VALUE

Investors expect to receive a return on their investments in the form of interest or other returns. We need to increase the original amount invested by adding the interest that accrued during the time of the investment. *Future value is the value in rands that an investment or series of investments will grow to over a stated time period at a specified interest rate.* The following notation will form the basis of the formulae to be applied in this chapter:

FV: The amount of cash which will have accrued by a given date resulting from earlier single sum or periodic investments.

PV: The value of an investment at the beginning of a period, sometimes referred to as the *principal sum*.

r : The interest rate.

PMT: The periodic investments or instalments made, excluding single lump-sum investments. This may occur at the end or beginning of each period.

n : The number of periods for which the investment is to receive interest.

Single amount, single period

Where a single amount is invested for a period of one year, the calculations are relatively straightforward.

Example 2.1: Future value: one year hence

An amount of R100 is invested for one year at a rate of 12% p.a. What is the future value of this investment at the end of the year?

$$\begin{aligned} \text{FV} &= \text{PV} (1 + r) \\ &= \text{R}100 (1.12) = \text{R}112 \end{aligned} \quad (\text{Formula 2.1})$$

The interest on the investment is clearly R12, and the future value of the investment after one year is R112.

Single amount, multiple periods, annual interest compounded

An investment of a single amount accrues interest in each period and, frequently, the interest is added to the original investment rather than being paid out. This is referred to as *compound interest*. This very important concept in financial management results in interest being earned on interest. The compounding effect has an immense impact on the value of an investment, especially if the period of investment is lengthy. For

example, R100 invested at 15% for three years, with interest compounded annually, amounts to R152, while if it was left for 30 years, it would accumulate to R6 621 and if left for 50 years would accumulate to R108 366. If interest was not compounded in this way, the interest every year would remain at R15 and the investment would grow by this amount each year and amount to R550 after 30 years or R850 after 50 years. Interest is not earned on interest. This is referred to as *simple interest*.

Example 2.2: Calculating the future value: more than one year

An amount of R100 is invested for 10 years at a rate of 12% p.a. compound interest. What is the future value of this investment at the end of 10 years?

Using Formula 2.1 we know that the future value at the end of year one is R112. If this whole amount remains invested for another year, the future value at the end of year two will be:

$$FV = R112 (1.12) = R125.44$$

The interest for the second year is: Page 2-3

$$R125.44 - R112.00 = R13.44$$

The interest for the second year is R1.44 higher than for the first year. This is because interest at the rate of 12% has been earned on the interest of R12 earned at the end of the first year as well as on the original R100. This can be checked by calculating the interest for one year on R12 at 12% p.a.

$$\begin{aligned} &= R12 \times 0.12 \times 1 \\ &= R1.44 \end{aligned}$$

In order to develop a formula for compound interest, the following logic can be applied:

For year one	$FV = R100 (1.12) = R112$
For year two	$FV = R112 (1.12) = R125.44$
For year three	$FV = R125.44 (1.12) = R140.50$

For year two the calculation could be expressed as:

$$FV = R100(1.12)(1.12)$$

For year three the calculation could be expressed as:

$$FV = R100(1.12)(1.12)(1.12)$$

This can be generalised to:

$$FV = PV (1 + r)^n \quad (Formula 2.2)$$

For the example above this is:

$$\begin{aligned} FV &= R100 (1.12)^{10} \\ &= R100 \times 3.1058 \\ &= R310.58 \end{aligned}$$

It is quite easy to calculate $(1.12)^{10}$. We can also use financial calculators, Excel spreadsheets or Table A. Table A, which is given at the back of this text, is referred to as the *future value of R1* table. It is drafted for all the commonly used interest rates for up to 50 different periods. By consulting Table A and looking for the 12% column and the 10-period row, the figure 3.1058 will be located. This is multiplied by the principal sum of R100 in order to arrive at the future value of R310.58.

If only the interest component is required, the principal sum can be deducted. In this case the compound interest earned for the 10 years is R310.58 less R100, that is R210.58.

We now have a formula for determining the future value and, we can calculate any missing variable. The effect of compounding on the value of an investment can be remarkable. The following graph depicts the future value of an investment at an interest rate of 12%, assuming both compound interest and simple interest. The difference is due to interest being earned on interest. **Page 2-4**

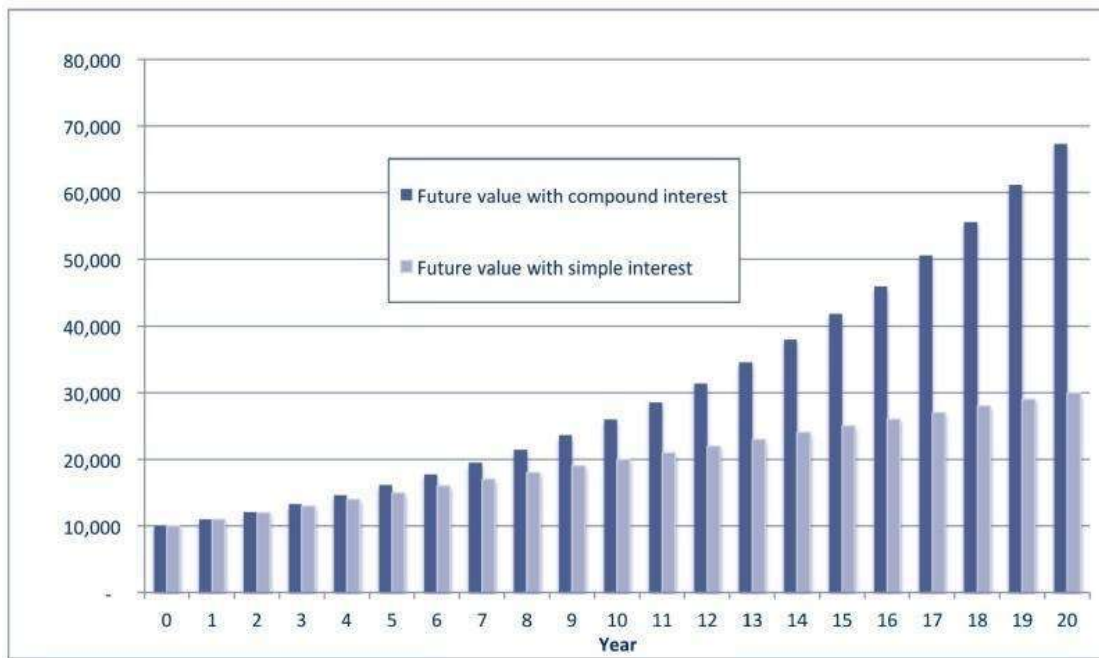


Figure 2.1 Future values – compound interest and simple interest

Assume that you have always wanted to be a millionaire. Well, if you invest R10 000 today at an interest rate of 10% for 50 years, you will reach your target of making over a million rand. Of course, a million rand will mean less in 50 years' time than it does today.

$$FV = R10\,000 (1 + 0.10)^{50} = R1.174m$$

The level of interest rates or returns also has a significant effect on the future value. The graph below indicates the effect that compounding at different interest rates has on future value.

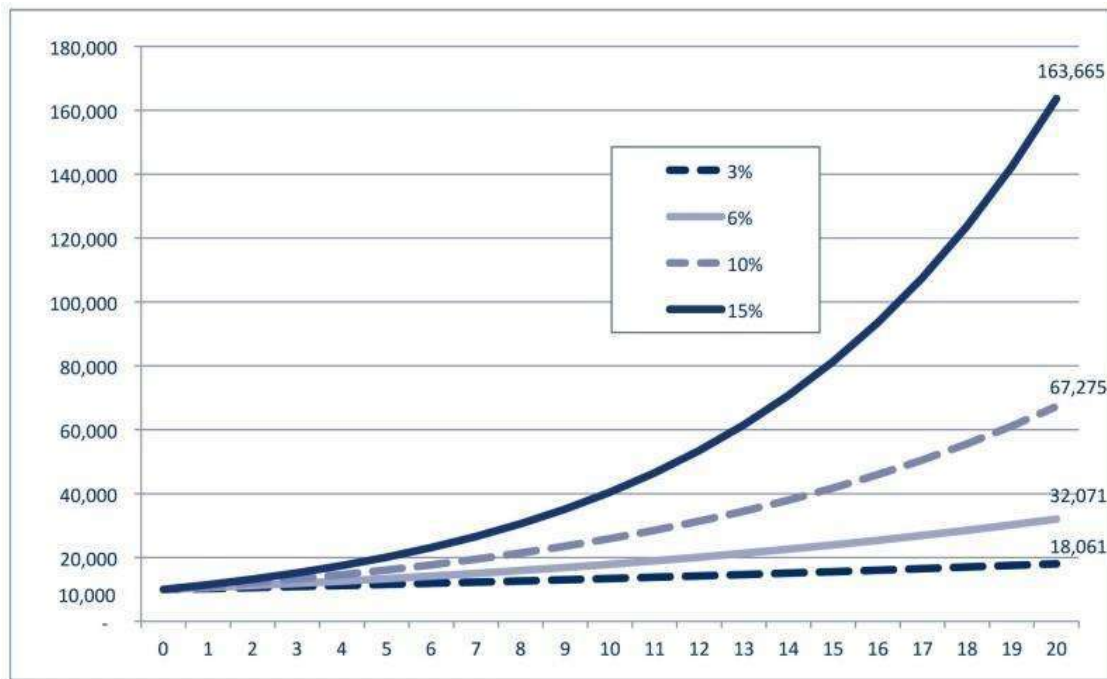


Figure 2.2 Future value of single investment of R10 000 at varying interest rates

Warren Buffett is one of the world's wealthiest men and the world's most famous investor, worth over \$109.7 billion (about R1 982 billion) in 2023. Investing in his company, Berkshire Hathaway from the beginning of 1965 to the end of 2022 would have resulted in a compound return of 19.8% per year. Assume that an old uncle invested \$1 000 on your behalf in Berkshire Hathaway at the time. What would you have been worth at the end of 2022? Remember we are talking about a \$1 000 investment 58 years ago. We can use Formula 2.2 as follows: [Page 2-5](#)

$$FV = 1\,000 (1+0.198)^{58} = \$35\,521\,839$$

Imagine the growth in wealth from \$1 000 to about \$35.5 million. You could retire before starting to work. Perhaps, this is why Albert Einstein stated that the most powerful force in the universe is compound interest. However, assume that your uncle had rather invested the \$1 000 in the general share market (S&P500). This would have resulted in a compound return of 9.9% per year. What is the accumulated sum at the end of 2022? Applying Formula 2.2 again:

$$FV = 1\,000 (1+0.099)^{58} = \$238\,708$$

What a difference 9.9% (0.198 – 0.099) per year makes! Over a long period, even a few percentage points difference add up to a significant amount due to the impact of compound interest. Of course this is in the USA and we would like to know if you

could have done as well in South Africa. We will come back to this question a little later in the chapter.

Example 2.3: Calculating the principal amount

An investor wishes to invest a sum of money which will accumulate to R310.58 in 10 years' time. How much must be invested today, if the investor can earn a rate of 12% per year?

Changing the subject of Formula 2.2, it can be stated as:

$$\begin{aligned} PV &= \frac{FV}{(1 + r)^n} && \text{(Formula 2.3)} \\ &= \frac{310.58}{(1.12)^{10}} \\ &= \frac{310.58}{3.1058} \\ &= R100 \end{aligned}$$

As expected from the results of Example 2.2, the required investment is R100. Table A may also be used to expedite the calculation of $(1.12)^{10}$. Looking at Table A, the future value factor is 3.1058. We can also use a financial calculator. We solve the same problem later in the chapter using a financial calculator – see [Section 4](#).

Example 2.4: Calculating the number of periods

An investor is informed that an investment of R100 will grow to R310.58. If it is known that the applied rate is 12%, how many periods are necessary for the R100 to become R310.58?

Developing a formula to solve this would require the use of logarithms. However, by referring to Table A, and finding the relevant factor, the number of periods can be determined by inspection as follows: Page 2-6

$$\begin{aligned} (1 + r)^n &= \frac{FV}{PV} \text{ or } n = \frac{[(\text{LN } FV - \text{LN } PV)]}{\text{LN } (1 + r)} && \text{(Formula 2.4)} \\ (1.12)^n &= \frac{310.58}{100} \\ &= 3.1058 \end{aligned}$$

The number 3.1058 is the factor defined in Table A. Because it is known that the interest rate is 12%, it is possible to move down the 12% column until the number

nearest to 3.1058 is found. In this example it is found in the 10-period row. Let's go back to the formula:

$$(1.12)^n = 3.1058$$

We can use natural logarithms to solve for n :

$$\begin{aligned} n \ln 1.12 &= \ln 3.1058 \\ n &= \ln 3.1058 / \ln 1.12 \\ n &= 1.1333 / 0.11333 \\ n &= 10 \text{ years} \end{aligned}$$

Example 2.5: Calculating the interest rate

An investor is given the opportunity of investing R100 today with a promised future value of R310.58 in 10 years' time. At what rate is the investment accruing interest?

From Formula 2.4, it is possible to make r the subject of the formula as follows:

$$\begin{aligned} (1 + r)^n &= \frac{FV}{PV} && \text{(Formula 2.5)} \\ r &= (FV/PV)^{(1/n)} - 1 \\ &= (310.58/100)^{(1/10)} - 1 \\ &= 0.12 \text{ or } 12\% \end{aligned}$$

Instead of using the formula, Table A can again be used. This time one would search for a number close to 3.1058 by looking along the 10-period row. Once the closest number to 3.1058 is located, the column in which it is situated is the required interest rate. We can also use a financial calculator or Excel spreadsheets to determine the interest rate (See [Section 4](#) on financial calculators and Excel).

We analysed the performance of investing with Buffett in the USA. What about investing in South Africa? Allan Gray is one of South Africa's largest asset management companies. Allan Gray stated in 2023 that if you had invested R10 000 in June 1974 with Allan Gray then this investment would have grown to an incredible R317.5 million by 31 December 2022. If you had invested in the JSE All Share Index over the same period, your R10 000 would have grown to R14.2 million. Whilst the JSE appears to have offered a very good return, Allan Gray's performance seems to have been exceptional. What is the annual compound return that Allan Gray earned over the period? What is the annual compound return from investing in the JSE over the same period?

We can work with Formula 2.5 and we will count this as 48 years.

$$\text{Allan Gray: } R10\,000 (1+r)^{48} = R317.5\text{m}$$
$$r = (317\,500\,000/10\,000)^{(1/48)} - 1 = 24.104\% \text{ per year}$$

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$$\text{JSE All Share Index: } R10\,000 (1+r)^{48} = R14.2\text{m}$$
$$r = (14\,200\,000/10\,000)^{(1/48)} - 1 = 16.325\% \text{ per year}$$

In fact, Allan Gray has outperformed the JSE All Share Index by 7.78% per year and we can see what a big difference this can have on the final accumulated amount. Allan Gray's investment performance compares well to Warren Buffett's performance although we need to be careful as relative performance depends on the beginning and ending period.

Single amount, multiple periods, non-annual compounding

It often happens that an investment offers a rate of interest per annum, but stipulates that interest will be compounded at different intervals – quarterly or monthly, for example. This means that interest is added more frequently than once a year. As a result there is more opportunity for earning interest on interest. The net effect is that a higher return is obtained on the investment than would have been the case had interest only been compounded annually. In such a case the quoted per annum rate is called the *quoted* rate while the true return in annual terms is higher if the number of compounding periods is more than once a year, and is called the *annual effective* rate (see Formula 2.7).

Example 2.6: Future value, interest compounded monthly

An investor deposits R100 into an account which offers 12% p.a. interest compounded monthly. Find the value of the investment at the end of one year.

Because interest is compounded monthly, it can be said that 12% over 12 months – i.e. 1% interest – is added every month. Using Formula 2.2 it can be seen that the investment at the end of the second month would be:

$$FV = R100 (1.01) (1.01)$$

At the end of the twelfth month it would be:

$$\begin{aligned}
 FV &= R100 (1.01)^{12} \\
 &= R100 \times 1.1268 \\
 &= R112.68
 \end{aligned}$$

The extra 68 cents, although not significant over a short period, will compound over longer periods and can make a considerable difference. The effective rate in this example is 12.68%.

The formula required to generalise this calculation is as follows:

$$FV = PV \left(1 + \frac{r}{m}\right)^{mn} \quad (\text{Formula 2.6})$$

where m is the number of times within a year that interest is compounded and n is the number of years.

If the investment were made for 10 years, the future value would be:

$$\begin{aligned}
 FV &= R100 \left(1 + \frac{0.12}{12}\right)^{12 \times 10} \\
 &= R100 \times 1.01^{120} \\
 &= R100 \times 3.300 \\
 &= R330
 \end{aligned}$$

Page 2-8 It is clear that Table A can still be used. The difference is that the necessary adjustment must be made to the rate and the periods in order to read the table correctly. The formulae and the tables work in *periods* rather than years. Monthly interest in this case is 1% for 120 periods. Unfortunately, few tables extend as far as 120 periods. Rather use the formula or a financial calculator.

Annual effective rate

It is useful to be able to convert a quoted or nominal interest rate into an effective interest rate for many reasons, particularly in order to compare investment alternatives with different compounding periods. We can accept that two interest rates are equivalent if they have the same effect, that is, if a single amount invested at one rate and compounded annually for a length of time accumulates to the same future value as another rate compounded more frequently.

The effective annual rate of interest is the annual rate that if compounded once a year would give us the same result as the interest per period compounded a number of times per year. The effective annual rate is calculated as follows:

$$1 + r_e = \left(1 + \frac{r_n}{m}\right)^m$$

where r_e is the effective rate and r_n the nominal rate, compounded m times annually. This converts to:

$$r_e = \left(1 + \frac{r_n}{m}\right)^m - 1 \quad (\text{Formula 2.7})$$

Example 2.7: Converting nominal to effective interest rate

Ozbank offers the Ozplan Premium account, an investment account which requires a minimum investment of R10 000. The account offers an interest rate of 6% per year, interest compounded monthly. What is the annual effective interest rate?

Using Formula 2.7:

$$\begin{aligned} \text{Effective rate} &= \left(1 + \frac{0.06}{12}\right)^{12} - 1 \\ &= 1.005^{12} - 1 \\ &= 1.06168 - 1 \\ &= 6.1678\% \end{aligned}$$

This means that 6% per year, with interest compounded monthly is equivalent to 6.168% per year, with interest compounded annually.

Table 2.1 Effective rates

Compounding period	Number of compounding periods	Quoted rate	Effective rate
Annual	1	6.00%	6.0000%
Semi-annually	2	6.00%	6.0900%
Quarterly	4	6.00%	6.1364%
Monthly	12	6.00%	6.1678%
Daily	365	6.00%	6.1831%
Continuous e^r	2.71828	6.00%	6.1837%

Note: As m approaches infinity, we can use $e=2.71828$ as the base.

Page 2-9 We can use effective rates to compare investments or loans which have the same quoted rate but different compounding periods. In terms of “truth in lending” rules, banks may disclose the average percentage yield (APY) which is similar to computing the effective rate. Investec in May 2018 was advertising a 12-month fixed deposit rate of 7.53% nominal and Investec indicated that this represented an annual effective rate of 7.80%. The deposit earns interest on a monthly basis, so there are 12 compounding periods in a year. Is the advertised effective rate correct? If we apply

Formula 2.7, we can see that the effective rate is correctly stated at 7.80% $[(1+0.0753/12)^{12} - 1]$.

Continuous compounding

So far, in considering an investment which earns interest, we have used annual, semi-annual, quarterly, monthly or daily compounding. However, why stop there? We could invest so that interest is earned on interest every hour or every second. What happens is that we approach a limit. We can see from [Table 2.1](#) that the incremental increases in the effective interest rate gets smaller as we move from annual to daily compounding. If we extend [Table 2.1](#), then hourly compounding would represent adding interest of 0.000006849 $[6\%/(365 \times 24)]$ of the principal every hour. This would result in an effective rate of 6.18363%. As we approach infinity in terms of the number of compounding periods, the effective interest rate is $e^r - 1$, whereby e is equal to 2.71828. In our example, if the investment is offering continuous compounding, then this would result in an annual effective interest rate of:

$$\text{Annual Effective Rate} = 2.71828^{0.06} - 1 = 6.18365\%$$

The future value of an investment earning interest at 6% per year, with interest compounded continuously, is determined by multiplying the present value by e^m . If we were investing R10 000 at 6% for 10 years, with interest compounded continuously, then the future value would be:

$$\text{Future Value} = \text{Present Value} \times 2.71828^{(0.06 \times 10)}$$

$$\text{FV} = \text{R10 000} \times 1.822119 = \text{R18 221.19}$$

Alternatively, using the effective annual interest rate:

$$\text{FV} = \text{R10 000} \times 1.0618365^{10} = \text{R18 221.19}$$

Apart from simply determining the future value of an investment, continuous compounding (and discounting) has other uses in finance, such as in the pricing of options.

Interpolation

When using Table A to determine the rate or number of periods, the exact table reading, referred to as the *accumulation factor*, is seldom found. In order to obtain

more accurate solutions, interpolation should be used. Although not strictly arithmetically accurate, interpolation does provide a better answer than estimation.

Example 2.8: Finding the interest rate

An investor deposits R100 into an account which promises a future value of R154 in six years' time. At what rate is interest being compounded annually? **Page 2-10**

Using Formula 2.4:

$$(1 + r)^n = \frac{FV}{PV}$$
$$= \frac{154}{100} = 1.540$$

As we know it is for 6 years, by reading across Table A under six periods, we see that a factor of 1.54 lies between 7% and 8% for 6 years:

Periods	7%	8%
6	1.5007	1.5869

Note that the future value of 1.540 lies between 7% and 8%. The difference in the table readings between 7% and 8% is 0.0862 (1.5869 – 1.5007).

The factor of 1.54 required is 0.0393 (1.54 – 1.5007) above 7%. In order to determine how far past 7% it is using interpolation, the following calculation is used:

$$\frac{0.0393}{0.0862} \times 1\% = 0.456\%$$

The interest rate so calculated is 7.456% (7% + 0.456%)

Although interpolation may be useful, it is no longer necessary in a world of financial calculators and Excel spreadsheets [Financial calculator result: 7.4616%].

Series of investments, ordinary annuity (FVA) – multiple investments and multiple periods

So far we have dealt with a single investment at the beginning of a period and considered all the variables that can be determined. Investment opportunities are also available that require a series of payments of a fixed amount for a specific number of periods. These are known as *annuities*. For reasons which relate more to consistency of formulation and generally accepted usage than to practical investment procedure, we will assume that the first instalment is generally invested at the *end of the period*.

This is known as an *ordinary annuity* and we will apply this concept throughout this text unless otherwise stated. We will now develop the necessary formulae and tables.

Example 2.9: Calculating the future value of an ordinary annuity

An investor pays equal instalments of R100 at the end of each year into a savings account yielding an interest rate of 12% per year, compounded annually. What is the future value of the investment at the end of three years?

An ordinary annuity means that the payment occurs at the end of each period and the following table depicts the cash flows and the investment of each annuity payment.

Table 2.2 Future value of an ordinary annuity (FVA) at 12%

Year end	0	1	2	3
Instalment		R100	R100	R100.00
			$(1.12)^1$	R112.00
			$(1.12)^2$	R125.44
			Future value =	<u>R337.44</u>

Page 2-11 The first instalment earns interest for two years, the second instalment earns interest for one year, and the third instalment receives no interest as it is deposited at the end of Year 3. This can be expressed as:

$$\begin{aligned} \text{FVA} &= \text{R}100 \times 1.12^2 + \text{R}100 \times 1.12^1 + \text{R}100 \times 1.12^0 \\ &= 337.44 \end{aligned}$$

If the instalment is taken as the common factor, the future value interest factor of an annuity can be written as:

$$\text{FVIFA} = \sum_{t=1}^3 (1.12)^{3-t}$$

Where t is the period for each instalment, this translates into the following formula:

$$\text{FVIFA} = \sum_{t=1}^n (1 + r)^{n-t}$$

This formula is cumbersome and requires a separate calculation for each of the years. A more workable formula has been developed which achieves the same result. It is

particularly useful for calculations requiring fractions of rates or periods not provided for in the tables.

$$\begin{aligned}
 \text{FVA} &= \text{PMT} \times \left[\frac{(1 + r)^n - 1}{r} \right] && (\text{Formula 2.8}) \\
 &= \text{R}100 \times \left[\frac{(1 + 0.12)^3 - 1}{0.12} \right] \\
 &= \text{R}100 \times 3.3744 \\
 &= \text{R}337.44
 \end{aligned}$$

We can also use Table B to determine the future value of an ordinary annuity. This table is known as the *future value of an annuity of R1 per period* and contains the future value interest factor of an annuity (FVIFA). Table B is closely related to Table A. In fact, Table B is the accumulation of Table A for $n - 1$ periods plus a factor of one. This relationship exists as Table B assumes the payment takes place at the end of the period, while Table A assumes the payment occurs at the beginning.

n	Table A – 1%		Table B – 1%
1	1.0100		1.0000
2	1.0201	e.g.	2.0100
3	1.0303	+1	3.0301
4	1.0406		4.0604
5	1.0510		5.1010

The FVIFA is the portion of Formula 2.8 which appears in brackets. Using Table B to calculate the FVA therefore requires only the product of the instalment and the Table B FVIFA which for our example is as follows:

$$\begin{aligned}
 \text{FVA} &= \text{PMT} \times \text{FVIFA}_{n,r} \\
 &= \text{R}100 \times \text{FVIFA}_{3,12\%} \\
 &= \text{R}100 \times 3.3744 \\
 &= \text{R}337.44
 \end{aligned}$$

Page 2-12 Similar principles to those already developed for dealing with a single amount investment can be used to determine the rate or number of periods as well as for more frequent compounding.

Series of investments, annuity due

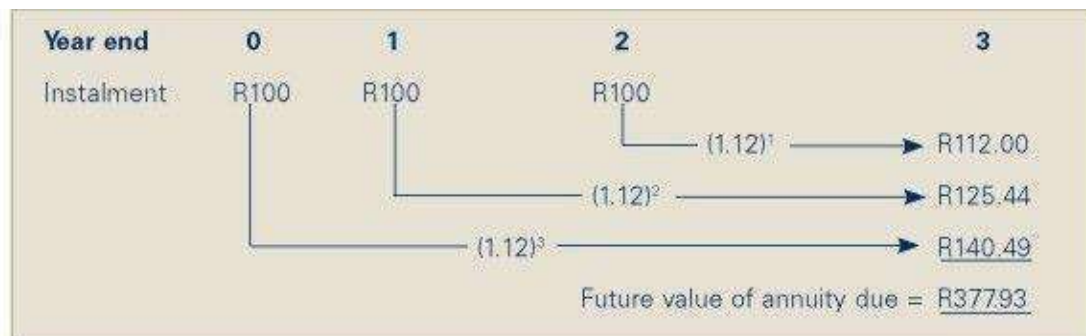
Where specifically stated, an annuity may be structured by depositing the first instalment at the beginning of the period. Such an annuity is known as an annuity *due*.

Because Table B has been compiled for ordinary annuities, certain adjustments need to be made if an annuity due accumulation factor is required. In our example, we can have a three year annuity of an equal payment which occurs at the end of each period, or an equal payment which occurs at the beginning of each period.

Example 2.10: Calculating the future value of an annuity due

An investor pays an instalment of R100 at the beginning of each year into a savings account yielding 12% per year interest compounded annually. What is the future value of the investment at the end of three years? It is useful to use a diagram of cash flows to develop the relevant formula.

Table 2.3 Future value of an annuity due at 12%



In this case all three instalments earn interest. We could undertake three separate calculations and sum the results. In order to adjust the Table B reading, an annuity due requires the reading under $3 + 1$ periods, but without the first instalment. The table reading is thus adjusted by adding one period, that is, by reading under four periods rather than three periods and then deducting one from the factor indicated by Table B. This converts the table reading to reflect investment at the beginning of a period rather than the end of the period as follows:

$$\begin{aligned}
 FVA &= PMT \times (FVIFA_{(3+1); 12\%} - 1) \\
 &= R100 \times (4.7793 - 1) \\
 &= R377.93
 \end{aligned}$$

The formula for the future value of an ordinary annuity, Formula 2.8, can similarly be adjusted for an annuity due as follows:

$$\begin{aligned}
 FVA_{\text{due}} &= PMT \left[\left(\frac{(1+r)^{n+1} - 1}{r} \right) - 1 \right] && \text{(Formula 2.9)} \\
 &= R100 \left[\left(\frac{(1+0.12)^4 - 1}{0.12} \right) - 1 \right] \\
 &= R377.93
 \end{aligned}$$

It is possible to multiply our solution from applying Formula 2.8 by $(1+r/m)$ to arrive at the same solution¹. In our example (see previous page), $R337.44 \times (1+0.12/1) = R377.93$.

Page 2-13 An annuity due will have a higher future value than an ordinary annuity because the investment is made earlier, which means that each instalment will earn interest for an additional period.

Future values when the timing of the cash flows and the compounding periods differ

So far, to use the formulae, and financial calculators, we have had a cash flow per period and an interest rate for the same period. Therefore, if there are quarterly payments, interest is compounded quarterly. What do we do when there may be quarterly payments but interest is compounded monthly? For example, a company deposits R10 million per quarter for 5 years and interest is 6% per year, compounded on a monthly basis. One way is to determine the effective interest per quarter and apply this in our calculations of future value. The monthly rate is 0.5% [$6\%/12$] and there are 20 quarters.

$$\begin{aligned}\text{Effective quarterly rate} &= (1.005)^3 - 1 = 0.015075 \\ \text{FVA} &= 10\text{m} \times \left[\frac{(1 + .015075)^{20} - 1}{0.015075} \right] \\ &= R10\text{m} \times 23.14075 \\ &= R231.41\text{m}\end{aligned}$$

2 PRESENT VALUES

Present value is a powerful concept in finance. If you receive money today, you can invest it to earn interest. This means that you will prefer to receive R100 today rather than R100 in a year's time, as you can invest it and earn a return. Up to now we have wanted to know what the future value will be in, say, five years' time if we invest, say, R100 today. For example, R100 invested for one year at 8% will result in a future value of R108. We now turn the question around: how much do we need to invest today to reach a target of R108 if the interest rate is 8%?

The present value, that is the value today, of a stream of expected future cash flows can be established in much the same way as future values. Many valuation problems in financial management involve calculating the present value of a stream of cash

flows that is expected in the future. Present value enables us to make direct comparisons between investments. It is difficult, for example, to decide between alternative investments offering income streams which may differ in the level of cash flows and the timing of such cash flows. For example, how do we compare investments A and B with cash flows as set out in the following table? We can compare the investments by determining the present value of the cash flows of each project, but more about that later.

Project	0	1	2	3
Cash flows in Rm				
A	-20	16	5	5
B	-20	1	5	23

The calculation of present values uses similar reasoning to the calculation of future values. Instead of referring to *compounding*, we refer to *discounting*, which is the inverse of compounding. The rate used to move cash flows back to the present is called the discount rate. Why call this a discount rate? For example, if we purchase goods at a discount, we pay less than 100% (1) of the price. You will notice that the factors in Table C are all less than 1 and this confirms that we are discounting back to today's value. **Page 2-14**

Single amount, single period, annual discounting

A rand today is worth more than a rand at a future time. This is the essence of the concept of time value. Money received immediately is likely to be used productively and to be worth more in the future.

Different individuals will generally have differing time-preference rates for money – that is, they will apply different discount rates to future cash flows. These differences can be attributed to the different needs or opportunities which would require them to make use of the cash. The primary factors for determining an individual's time-preference rate for money, remain the *time value of money*, the *risk attached to the investment*, and *inflationary expectations*.

Example 2.11: Calculating the present value of a future amount due in one year's time

An investment offers the opportunity to receive R100 one year from now if R90 is paid immediately. Should an investor who applies a 12% discount rate make the investment?

As present value is the inverse of future value, Formula 2.3 can be applied.

$$\begin{aligned} PV &= \frac{FV}{(1 + r)^n} \\ &= \frac{100}{(1.12)^1} \\ &= R89.29 \end{aligned}$$

The present value of the investment to the investor is R89.29. As this is less than the cost of the investment, of R90, it is not a worthwhile investment. One way of conceptualising this outcome is to ask whether the investor would give R90 (the cost of the investment today) in exchange for R89.29 (the value of the investment today). The answer is obviously no. In much the same way as with future values, the formula can be adapted in order to determine other variables if they are unknown.

Single amount, multiple periods, annual discounting

An amount of money to be received more than one year in the future clearly has an even smaller present value than if it were to be received in a year's time. Because interest rates are compounded annually, the same principle is applied to discounting.

Example 2.12: Calculating the present value of a future amount due more than one year in the future

An investment offers the opportunity to receive R100 in ten years' time. If an investor applies an interest factor of 12%, what is the highest price which will be offered for the investment?

Applying Formula 2.3 the outcome is as follows:

$$\begin{aligned} PV &= \frac{FV}{(1 + r)^n} \\ &= \frac{100}{(1.12)^{10}} \\ &= R32.20 \end{aligned}$$

The investor would therefore be prepared to pay any amount up to R32.20 for the investment. If the required payment was R30, for example, the investor would purchase the investment because it would be equivalent to paying R30 today and receiving R32.20 today. **Page 2-15**

Formula 2.3 can also be written as:

$$\begin{aligned}
 PV &= FV \times \left[\frac{1}{(1+r)^n} \right] \\
 &= R100 \times \left[\frac{1}{(1.12)^{10}} \right] \\
 &= R100 \times 0.3220 \\
 &= R32.20
 \end{aligned}$$

Table C reflects discount factors for all commonly used ranges of interest rates and periods. It is referred to as the *present value of R1* table, or PVIF, denoting that it provides the present-value interest factor for a single sum to be received at a future date. It displays the portion of the formula which is in brackets above.

Stream of cash flows, ordinary annuity (PVA)

A series of equal cash flows, the first of which is due at the end of the first period, is called an *ordinary annuity*. In order to establish the present value of an ordinary annuity (PVA), the future cash flows must be discounted to the present.

Example 2.13: Calculating the present value of an ordinary annuity

An investor wants to buy an annuity of R100 for the next three years. If a bank is prepared to pay interest at 12%, how much must be invested today?

This is another way of asking what the present value of a stream of three annual payments is. It can be illustrated as shown in [Table 2.4](#).

Table 2.4 Present value of an ordinary annuity at 12%

Year end	0	1	2	3
Cash flow		R100	R100	R100
		$\frac{1}{(1.12)^1}$	$\frac{1}{(1.12)^2}$	$\frac{1}{(1.12)^3}$
	R89.28			
	R79.72			
	R71.18			
	<u>R240.18</u>			
	= Present value of the annuity			

From [Table 2.4](#) it is apparent that:

$$\begin{aligned}
 PVA &= R100 \left(\frac{1}{1.12^1} + \frac{1}{1.12^2} + \frac{1}{1.12^3} \right) \\
 &= R100 \sum_{t=1}^n \left(\frac{1}{1.12} \right)^t
 \end{aligned}$$

This relationship can be generalised and expressed in the equation:

$$PVA = PMT \sum_{t=1}^n \left(\frac{1}{1+r} \right)^t$$

Page 2-16 It can be more conveniently expressed as follows:

$$\begin{aligned}
 PVA &= PMT \times \left(\frac{1 - \frac{1}{(1+r)^n}}{r} \right) && \text{(Formula 2.10)} \\
 &= R100 \times \left(\frac{1 - \frac{1}{(1.12)^3}}{0.12} \right) \\
 &= R100 \times 2.4018 \\
 &= R240.18
 \end{aligned}$$

Again, the above formula is particularly useful when a specific rate or period cannot be found in the tables.

A set of tables, referred to as the *present value of an annuity of R1 per period*, has been compiled. Table D provides the present value interest factor for the annuity (PVIFA).

Reading from Table D, the solution should be:

$$\begin{aligned}
 PVA &= PMT \times PVIFA_{3;12\%} \\
 &= R100 \times 2.4018 \\
 &= R240.18
 \end{aligned}$$

We can also use a financial calculator to solve these problems and do so in a separate section later in this chapter.

In order to conceptualise what is happening in practice, follow [Table 2.5](#).

Table 2.5 Ordinary annuity

Ordinary annuity	0	1	2	3
Opening balance		240.18	169.00	89.28
Interest earned for the year at 12%		28.82	20.28	10.71
Capital at end of year	240.18	269.00	189.28	100.00
Withdrawal end of the year	-	(100.00)	(100.00)	(100.00)
Ending balance	240.18	169.00	89.28	(0.00)

An investment of R240.18 today earning an interest rate of 12% will enable an investor to withdraw an equal payment of R100 each year for three years. This is a very powerful concept in real world financial applications. For example, you may

wish to invest now to be able to withdraw a pension of R100 000 per year for three years. Equipment may cost R240 180 and a company borrows this amount and is required to repay the loan and interest in the form of annuity payments of R100 000 each year.

Stream of cash flows, annuity due

A stream of cash flows may be structured in such a way that the initial payment is received immediately rather than at the end of the first period. The calculation of the present value of an annuity structured in this way requires an adjustment to the tables. Because the first payment is received immediately, its present value is equal to its quoted value – that is, it does not need to be discounted. **Page 2-17**

Example 2.14: Calculating the present value of an annuity due at 12%

An investor wants to buy an annuity of R100 for the next three years, each instalment being paid at the beginning of the year. If a bank is prepared to pay interest at 12%, how much must be invested today?

It can be seen that the amount will be invested and the first instalment will immediately be repaid. It can be illustrated as shown in [Table 2.6](#).

Table 2.6 Calculating the present value of an annuity due at 12%

Year end	0	1	2	3
Cash flow	R100	R100	R100	
		$\frac{1}{(1.12)^1}$	$\frac{1}{(1.12)^2}$	
	R89.29			
	R79.72			
	<u>R269.01</u>			

How does an annuity due relate to an ordinary annuity? The following diagram reflects this relationship in terms of how we will restate the formulae and the tables.

	0	1	2
Annuity due	100	100	100
Ordinary annuity for 2 years	0	100	100
+			
Amount payable now	100		
	<u>100</u>	100	100

If Table D is used it will be noted that the reading is exactly the same as if it were a two- year annuity, but we need to add one to the present value factor because the immediate instalment is not discounted. Using Table D:

$$\begin{aligned}
 PVA_{\text{due}} &= PMT \times (PVIFA_{n-1; 12\%} + 1) \\
 &= R100 \times (PVIFA_{2; 12\%} + 1) \\
 &= R100 \times (1.6901 + 1) \\
 &= R100 \times 2.6901 \\
 &= R269.01
 \end{aligned}$$

The present value of R269.01 for an annuity due can be compared with that of an ordinary annuity calculated in Example 2.13 with a present value of R240.18. The annuity due has a higher present value because the cash flows occur earlier than in the case of an ordinary annuity.

Formula 2.10 can be adjusted for annuities due as follows:

$$\begin{aligned}
 PVA_{\text{due}} &= PMT \times \left\{ \left[\frac{1 - \left(\frac{1}{(1+r)^{n-1}} \right)}{r} \right] + 1 \right\} && \text{(Formula 2.10a)} \\
 &= 100 \times \left\{ \left[\frac{1 - \left(\frac{1}{(1.12)^2} \right)}{0.12} \right] + 1 \right\} \\
 &= R100 \times 2.69005 \\
 &= R269.01
 \end{aligned}$$

We can also multiply our solution to Formula 2.10 by (1+r/m) to obtain the same value, so that $R240.18 \times (1 + 0.12/1) = R269.01$. **Page 2-18**

What the formula is doing is reducing the number of discounting periods by one and effectively adding [1 x PMT] to recognise that the present value of the first payment equals its quoted value, as it occurs today.

In order to further conceptualise what is happening in practice, follow [Table 2.7](#).

Table 2.7 Annuity due

Annuity due	0	1	2
Opening balance		169.01	89.29
Interest earned for the year at 12%		20.28	10.71
Capital at end of year	269.01	189.29	100.00
Withdrawal end of the year	(100.00)	(100.00)	(100.00)
Ending balance	169.01	89.29	-

Stream of cash flows, deferred annuity

A *deferred annuity* is an annuity which begins at some time in the future. This would occur if an investor wanted to invest a sum of money now, but wanted the annuity to commence only at some future date. An example of a deferred annuity is a pension plan.

Example 2.15: Present value of a deferred annuity

Let's assume an investor wishes to invest a sum of money today which will yield three equal instalments of R100, the first payable three years from today. If interest accrues at 12%, what amount must be invested?

This is a present-value problem, requiring us to compute the value today of a stream of future cash flows, the first of which is deferred. The cash flows can be illustrated as shown in [Table 2.8](#).

Table 2.8 Present value of an annuity deferred at 12%

Year end	0	1	2	3	4	5
Cash flow				R100	R100	R100
			$\frac{1}{(1.12)^3}$			
	R71.18 ←					
			$\frac{1}{(1.12)^4}$			
	R63.55 ←					
				$\frac{1}{(1.12)^5}$		
	R56.74 ←					
	<u>R191.47</u>					

Table D is compiled on the assumption that the first cash flow arises at the end of period one. Because of the deferred period, the table readings can easily be adjusted as follows:

$$\begin{aligned} PVA &= 1 \times (PVIFA_{5, 12\%} - PVIFA_{2, 12\%}) \\ &= R100 \times (3.6048 - 1.6901) \\ &= R100 \times 1.9147 \\ &= R191.47 \end{aligned}$$

Page 2-19 The investment of R191.47 will earn interest for three years at the end of which the first instalment of R100 will be paid out. The balance will earn interest for another year before the next instalment is paid out, and similarly for the third instalment. The amount of R191.47 is the amount which must be invested at period 0 to make the cash flows possible. The following diagram indicates what is happening with a deferred annuity:



Figure 2.3 Deferred annuity

In the first three years, interest is earned until the end of Year 3 at a rate of 12%, when the first withdrawal of R100 occurs. In Year 4, interest is earned on the balance of R169 so that it will reach a level of R189 at the end of Year 4. After paying out another R100, there remains a balance of R89, which with interest will grow to R100 by the end of Year 5, which exactly equals the withdrawal of R100.

Uneven stream of cash flows

Companies will often be required to evaluate investments or projects which result in an uneven stream of future cash flows. Further, cash flows may be negative in certain periods. What do we do to determine present values in such cases? The annuity tables cannot be used. Each cash flow must be treated separately and discounted using Table C: *the present value of R1*. The present value of any cash flow can then be summed and the net present value of the stream of uneven cash flows established.

Let's go back to an earlier example. How do we evaluate Projects A and B?

Project	0	1	2	3
Cash flows in Rm				
A	-20	16	5	5
B	-20	1	5	23

Page 2-20 We are required to discount each cash flow. We will assume a discount rate of 10%. If we add up the cash flows, Project A results in total cash flows of R26m,

whilst Project B results in total cash flows of R29m. Yet we have to wait longer for the cash flows of B. Let's discount each cash flow by applying the formula $1/(1 + r)^n$ or by referring to Table C. Otherwise, you can also use a financial calculator.

Project	0	1	2	3
Cash flows in Rm				
A	-20	16	5	5
PV Factor		0.9091	0.8264	0.7513
		14.5455	4.1322	3.7566
Present Value		22.4343		

If we do the same for Project B, the present value is R22.32m. This means that although Project B has higher total cash flows, Project A has a higher present value of cash flows. As the cost for both projects is equal, we would prefer to select Project A. We will use a financial calculator to solve this problem later in the chapter.

Perpetuities

All annuities discussed so far have a finite life. The cash flows take place over a specific time and then cease. There are cases of annuities which provide cash flows for an infinite period. Such cash flows are called *perpetuities*. An example of perpetuity would be a non-redeemable preference share paying a fixed dividend.

Example 2.16: Calculating the present value of a perpetuity

An investor wants to buy 1 000 non-redeemable 9% preference shares of R1 each. If the interest rate which he applies is 12%, what is the present value of the investment?

In essence the investor is buying a future cash flow in perpetuity amounting to 9% of R1 000, that is R90. Because a 12% return on the investment is expected, this problem requires the principal sum to be determined.

$$\begin{aligned}
 PV &= \frac{PMT}{r} && \text{(Formula 2.11)} \\
 &= \frac{90}{0.12} \\
 &= R750
 \end{aligned}$$

The cash flow in perpetuity of R90 represents a return of 12% on R750. The investor who requires a return of 12% will therefore be prepared to pay no more than R750 which is the present value of the investment.

Let's relate the formula of perpetuity to that of an annuity. We determine the present value of an ordinary annuity by applying the following formula:

$$PVA = PMT \left[\frac{1 - \frac{1}{(1+r)^n}}{r} \right]$$

As n approaches infinity, $1/(1+r)^n$ approaches zero and the formula becomes $PMT \times 1/r$, which is the perpetuity formula. **Page 2-21**

Growing perpetuities

If a cash flow is growing at a constant rate, then we call this a growing perpetuity and this is a very useful concept in valuing companies, and so we will return to this in a later chapter in another guise. The following could be a future cash flow stream which is discounted at a given interest rate, r .

$$PV = \frac{PMT_1}{(1+r)} + \frac{PMT_2}{(1+r)^2} + \frac{PMT_3}{(1+r)^3} + \dots$$

If the payment is growing at a steady rate, then we can use the payment just made, PMT_0 , and apply the growth rate and so the formula becomes:

$$PV = \frac{PMT_0(1+g)}{(r-g)} \quad \text{(Formula 2.12)}$$

We will come back to this in a later chapter and will explain the derivation of this formula in greater detail.

If a company has just made a payment of R10 million and this is expected to grow at the expected inflation rate of 3% per year and the discount rate is 8%, then the present value of this payment stream is:

$$PV = \frac{10m(1 + 0.03)}{(0.08 - 0.03)} = \text{R206 million}$$

Growing annuity

If an amount is growing at a constant rate for a specified number of years, then we can use the following formula to determine the present value rather than discounting each cash flow.

$$\text{PV of a Growing Annuity} = \text{PMT} (1 + g) \times \left[\frac{1 - \frac{(1 + g)^n}{(1 + r)^n}}{r - g} \right] \quad (\text{Formula 2.13})$$

Assume that you have just graduated and you are expecting an annual salary of R360 000 (today's value) which you expect to grow at a rate of 6% per year for the next 40 years, at which time you will retire. What is the present value of your future salary over your working life if your required return is 9% per year? Assume an annual salary payable once a year. [You can easily change this by using monthly rates and your monthly salary.]

$$\text{PV of a future salary} = \text{R360 000} (1 + 0.06) \times \left[\frac{1 - \frac{(1.06)^{40}}{(1.09)^{40}}}{0.09 - 0.06} \right] = \text{R8 554 551}$$

This means that the present value of your future salary earned over 40 years is just over R8.5 million. This is before taking into account such issues as taxation which will reduce the real worth of your future salary. We are assuming that your first year's salary will be R381 600.

We can do these calculations the “long” way by using Excel spreadsheets. Invariably this is how it is done in practice and we will show you later how to use Excel spreadsheets to solve time value of money problems. **Page 2-22**

Inflation and real returns

The effect of inflation is to reduce the value of money over time. Whilst investing in a bank account may offer an interest rate of 4.8% per year, we need to consider the impact that inflation will have on the purchasing power of money. If expected inflation is 3% per year then you are not really making 4.8%. A R1 000 investment earning 4.8% will result in you accumulating R1 048 by the end of the year. However, you will need R1 030 at the end of the year for what you could have purchased for R1 000 at the beginning of the year. Obviously, this will affect pension decisions as we may think that a pension of R30 000 per month is sufficient but that is based on its current purchasing power. What will a pension of R30 000 per month starting in 30 years' time be able to purchase in relation to today's prices? In other words, what is the present value of R30 000 in 30 years' time? Inflation of 3% per year amounts to 0.25% per month.

$$PV = R30\,000 \times \left[\frac{1}{(1.0025)^{360}} \right]$$

$$PV = R30\,000 \times 0.4070 = R12\,210$$

This means that a pension of R30 000 in 30 years' time will be able to buy only what R12 210 can buy today, and that is at the beginning of the retirement period. It is worse at the end. The moral of the story is to include inflation in your investment calculations. A real return refers to the return we make after deducting inflation. If an investment is offering 4.8% per year and inflation is 3%, then the real return is approximately 1.8%. A more accurate way of calculating the real return is as follows:

$$1 + \text{real return} = (1 + \text{nominal return}) / (1 + \text{inflation rate})$$

We will come back to this again in a later chapter.

3 SOME REAL WORLD APPLICATIONS

Retirement planning

In South Africa, the combined effects of people living longer lives, low interest rates and lower expected future returns from the share market, mean that the economy will not be able to afford to pay old age pensions and retirees will not have accumulated sufficient amounts in their pension funds to retire comfortably. This issue is compounded by the low savings rates of South Africans. A few years ago, a *Business Day* editorial indicated that only 6% of South Africans would be able to retire without experiencing a reduction in their standard of living. Referring to an Alexander Forbes survey, the editorial stated that the average employee will retire on a pension equivalent to 28% of his/her final salary. Employees are contributing less than 10% of their salary to their pension fund whilst Old Mutual reported that employees should contribute 15% of their salary over 40 years in order to maintain their standard of living on retirement.

Time value of money principles are critical in retirement planning. Whilst the last thing on your mind may be retirement, it will affect your parents, and the earlier you start the better off you will be. Increasingly there are careers in financial planning and managing pension and retirement funds. So, how can we apply time value of money principles to retirement planning?

It is useful to be able to determine the single sum investment or the annuity required to provide adequately for retirement. To achieve this, an investor must estimate the

length of time prior to retirement, the return that will be earned on the funds invested during and after this period and, finally, the amount of funds required for retirement. **Page 2-23**

As most investors are not in a position to invest single lump sums, retirement plans are generally structured to require monthly contributions from the investor. On retirement, the investor will usually receive a pension which is in the form of an annuity.

Example 2.17: Calculating the required contribution to a retirement fund

Kate Smith wishes to retire in 30 years' time and has estimated that she will require a monthly pension income of R24 000 per month for 20 years subsequent to retirement. Kate will contribute to a retirement fund which will enable her to take out a monthly pension of R24 000 after retirement. The retirement fund is currently earning a return of 9% per annum, interest compounded monthly, and this level of return is expected to remain unchanged and to be sustainable over the next 50 years. Determine the monthly contribution that Kate is required to make to the retirement fund over the next 30 years.

First we need to calculate the present value required at retirement date to generate an annuity of R24 000 per month for 20 years. The interest rate is 0.75% per month (9%/12) and there are 240 months (12 X 20). The "present value" of the pension in 30 years' time is established as follows:

$$\begin{aligned} PV &= 24\,000 \times \left[\frac{1 - \frac{1}{(1.0075)^{240}}}{0.0075} \right] \\ &= 24\,000 \times 111.14495 \\ &= R2\,667\,479 \end{aligned}$$

This is the present value of the amount required to sustain a monthly payment of R24 000 for 20 years. This is the amount that she needs to accumulate in 30 years' time to be able to purchase an annuity or pension of R24 000 for 20 years after retirement.

The monthly contribution required over the next 30 years to generate an accumulated sum of R2 667 479 is calculated as follows:

$$R2\ 667\ 479 = PMT \times \left[\frac{(1.0075)^{360} - 1}{0.0075} \right]$$

$$= PMT \times 1\ 830.7435$$

$$PMT = R2\ 667\ 479 / 1830.7435 = R1\ 457.05 \text{ per month}$$

Of course, the above solution assumes that we are able to predict the return the fund will earn and the amount required at retirement age. In practice this is often very difficult as factors such as inflation and future returns are largely unknown. However, the advantage of such an exercise is that it gives us an indication, based on certain assumptions, of the level of investment required.

Let's see what happens when three very realistic events occur: people live longer, start saving later and the returns on retirement funds are expected to be at lower levels in the future. What happens to the required monthly payment if the expected lifespan after retirement is 30 years, returns have fallen to 6% per year and Kate starts saving only 20 years before retirement? Do the workings and you will find that Kate will need to have accumulated R4 002 999 by retirement date and this translates into a monthly contribution of R8 663.73 per month for the next 20 years. This is almost 6 times the figure of R1 457.05 with our previous assumptions. So it is easy to understand why there are corporate pension plans under water and unhappy retirees, although I am sure they are not complaining about living longer lives. Governments will find it difficult in the future to afford the payment of adequate old age pensions and the government is promoting later retirement. Increasingly we will see older **Page 2-24** people doing part-time work to supplement pensions. The pension crisis is on the horizon, so plan early. We will analyse retirement planning, inflation and living annuities in greater detail in Appendix 2.1.

Loan amortisation schedules

There are many real world applications of the repayment of loans on the basis of equal instalments over a number of years. What are some of the applications that will affect you now and in the future? Firstly, you may need to repay a student loan on the basis of instalments. You will purchase a car and may need to repay the car financing loan on an instalment basis. Also, you may purchase residential property which requires that you repay a mortgage loan over 30 years in monthly instalments.

Let's firstly assume the following example.

You have purchased an old but beautiful red Alfa Romeo for R100 000. The dealer has given you a car loan at an interest rate of 7% per year, interest compounded annually and repayable in equal instalments over 5 years. What is the annual instalment?

$$R100\ 000 = PMT \times \left[\frac{1 - \frac{1}{(1.07)^5}}{0.07} \right]$$

$$PMT = R100\ 000 / 4.1002$$

$$= R24\ 389.07 \text{ per year}$$

This can be broken down into payment of interest and principal and set out in a loan amortisation schedule as in the following table.

Table 2.9 Loan amortisation

Loan	100,000.00				
Interest	7%				
Term	5				
Year	Beg. Bal	Payment	Interest	Principal	End. Bal.
1	100,000.00	24,389.07	7,000.00	17,389.07	82,610.93
2	82,610.93	24,389.07	5,782.77	18,606.30	64,004.63
3	64,004.63	24,389.07	4,480.32	19,908.75	44,095.88
4	44,095.88	24,389.07	3,086.71	21,302.36	22,793.52
5	22,793.52	24,389.07	1,595.55	22,793.52	-
Total		121,945.35	21,945.35	100,000.00	

The first instalment of R24 389.07 is made up of R7 000 of interest ($R100\ 000 \times 7\%$) and a repayment of principal of R17 389.07. The balance at the end of the year is R82 610.93 ($R100\ 000 - R17\ 389.07$). The second instalment at the end of the second year includes interest of R5 782.77 ($R82\ 610.93 \times 7\%$) and a repayment of capital of R18 606.30. The interest component falls over time as the capital amount is reduced to zero by the end of the 5th year.

Mortgage loan

Assume that you have set your heart on an apartment that is 5 minutes from the beach and the price is R2 million. You have managed to obtain a 100% mortgage loan from the bank at an interest rate of 7.2%, interest compounded monthly, which means that you will be charged a monthly interest rate of 0.6%. The term of the loan is 30 years i.e. 360 months. What is your monthly repayment on the mortgage loan?

$$R2\,000\,000 = PMT \times \left[\frac{1 - \frac{1}{(1.006)^{360}}}{0.006} \right] \text{ so } PMT = R2\,000\,000/147.3214 \\ = R13\,575.76 \text{ per month}$$

If we draw a graph of the mortgage loan amortisation schedule over 30 years, it will indicate the balance of interest and the amortisation of the mortgage loan. The principal component is relatively small but increases over time. This can be quite disheartening in the first few years. For example, of the first month's instalment of R13 575.76, the interest component is R12 000 and only R1 575.76 will be allocated to repaying the R2 million loan. A monthly instalment of R13 575.76 over 30 years will result in the repayment of the R2 million loan and interest. So how much interest will be paid over the term of the mortgage loan?



Figure 2.4 Loan amortisation

If we multiply R13 575.76 by 360 months, we find that the total amount paid is close to R5 million (R4 887 274) and so total interest amounts to R2 887 274. That is quite a sum and so next time someone tells you that he purchased an apartment for R2 million and sold it for R5 million, the sums are a little more complicated than they seem. There are costs such as maintenance and you have to live somewhere so take off the rent you would have had to pay.

4 FINANCIAL CALCULATORS AND SPREADSHEETS

Using financial calculators

Financial calculators are programmed to compute future values and present values and other related variables. Financial calculators require you to enter the data using the relevant Input keys which for the HP-10BII are as follows:



What do these keys mean?

N = number of periods

I/YR = interest rate as a percentage

Enter a number, so if the rate is 10%, enter 10, not 0.10. Other calculators may reflect the interest rate per period as [i] and the number of periods as [n]. **Page 2-26**

PV = present value

PMT = annuity payment (Specify this as a zero when working with single sums only.)

FV = future value

In most cases, three or four inputs will be specified, and the financial calculator will solve for the remaining variable. On some calculators you will need to press the COMPUTE key prior to pressing the missing input key.

What to watch out for when using a financial calculator

Outflows are recorded as negative cash flows. An investment will be recorded in present value terms as a negative amount. If you borrow, you will receive the loan amount today which will be recorded as a positive cash flow and the repayments will be reflected as negative amounts. Ensure that you clear the memory. Enter a zero for any variable that is not relevant in a particular case. For example, when determining the future value of a single sum investment, enter a zero for the PMT key.

Ensure that the financial calculator is programmed for annual compounding by doing the following steps:

Enter 1, then press SHIFT key and then the PMT (P/YR) key. We will make adjustments later for non-annual compounding.

Let's do some of the previous examples by using an HP-10BII financial calculator.

An investment of R100 invested for 10 years earning 12% per year compound interest will result in a future value of R310.60. Key in the following input values and press the FV key for the solution.

N	I/YR	PV	PMT	FV
10	12	-100	0	310.6

First enter the present value as a negative number, -100 or press 100 followed by (-), depending on the calculator being used, and then press the PV key. Then enter 10 and press N, enter 12 and press I/YR, enter 0 and press PMT and then press FV (or Comp FV) to find the answer.

If given a FV of 310.6 with the number of periods being 10 and an interest rate of 12%, then key in the following variables and press the PV key to find the answer of -100:

N	I/YR	PV	PMT	FV
10	12	-100	0	310.6

What is the interest rate that will achieve a present value of R100 growing to R310.60 within 10 years? Enter the inputs, then press the [I/YR] key to determine the interest rate of 12%.

N	I/YR	PV	PMT	FV
10	12	-100	0	310.6

Page 2-27 We have assumed so far that the company has annual compounding. If the compounding period is less than a year, then we will need to input the number of compounding periods per year. This requires us to press the number of periods per year, then press the SHIFT key and (P/YR).

Let's go back to the first example and assume monthly compounding. We need to program the financial calculator for monthly compounding by undertaking the following steps:

Enter 12, and then press the SHIFT key and then the PMT (P/YR) key.

This means that the financial calculator is now programmed for monthly compounding. The present value is R100, the interest rate is 12% per year and N is 120 months.

N	I/YR	PV	PMT	FV
120	12	-100	0	330

Remember to input the annual interest rate. The financial calculator will divide it by the number of compounding periods per year. The effect of compounding interest each month rather than each year is to increase the future value from R310 to R330. Remember to input the number of months.

Present values

What is the present value of an annuity of R100 per year for three years at an interest rate of 12%?

N	I/YR	PV	PMT	FV
3	12	240.18	-100	0

The present value is R240.18. This assumes that the annuity occurs at the end of each year. What is the present value of an annuity that starts at the beginning of each year? Firstly, set the calculator to the beginning of year by pressing the SHIFT key and press the BEG/ END key. Then enter PMT of -100, N of 3, I/YR of 12, FV of 0 and then press PV to find the answer of R269.

N	I/YR	PV	PMT	FV
3	12	269	-100	0

Remember to reset your financial calculator back to end of period mode by pressing the SHIFT key and the BEG/END key. Usually we will assume that cash flows occur at the end of each period. Remember always to ensure that you clear the memory and all previous inputs by pressing the SHIFT key and then the C ALL key.

How do we use a financial calculator to determine the present value of an uneven stream of cash flows? We will use the CFj and the I/YR keys to input data and press the NPV key for the answer. Let's assume that Project A has the following cash flows:

YR	1	2	3
CF Rm	16	5	5

Assume there is no cash flow now today, i.e. time zero. You will still need to input a value, so enter zero. Firstly, input 0 then press the CFj key, then input 16 and press the

CFj key, input 5 and press the CFj key, input 5 and press the CFj key, input 10 and press the I/YR **Page 2-28** key and then press the SHIFT key followed by pressing the NPV key to find the answer of R22.4343m.

0 CFj 16 CFj 5 CFj 5 CFj 10 I/YR **SHIFT**
NPV
22.434

Using Excel spreadsheets

Although financial calculators can be very useful, in practice managers will prefer to use Excel spreadsheets to solve time value of money problems. Excel is a very powerful tool in many applications of corporate finance and useful in financial modelling. Like financial calculators, Excel has many built-in time value of money functions, yet it goes further as it allows us to obtain a visual perspective and permits us to put cash flows on a time line by using a cell to reflect a time period. It also allows us to build financial models so that we can see the effect of changing variables without redoing the whole calculation.

We place the relevant values in cells. Although this is not necessary as we can use Excel as a financial calculator, it does represent best practice as it enables us to undertake sensitivity analysis. Values are therefore anchored in cells and when we change the value in a particular cell the answer will change. We will use Excel to solve a few time value of money problems.

What is the future value of R10 000 invested today for 20 years at an interest rate of 6% interest compounded annually? In Excel we will place these values within cells. In our example there are no annuity payments and so the type (beginning of period =1, end of period =0) is not relevant. We have used Cell B8 to place the Future Value function which is; =FV(rate, nper, pmt, PV, type). We place the rate in Cell B6, and the number of periods is placed in Cell B5. The PV is placed in Cell B3. The payment (pmt) is placed in Cell B4 and type is placed in Cell B7.

	A	B
1	Future Value Calculations	
2		
3	Present Value	10,000.00
4	Payment	0.00
5	Periods	20
6	Rate	6%
7	Type	0
8	Future Value	32,071.35
9		
10	What is in Cell B8?	=FV(B6,B5,B4,-B3,B7)

The objective is to calculate the future value and so we type the Excel Future Value function in Cell B8, which in this case would be: =FV(B6,B5,B4,-B3,B7). The answer of R32 071 appears in the cell. What this means is that we can now change each variable and see what happens to future value. For example, if we expect that the interest rate will be 10%, then place this in Cell B6, and the answer in Cell B8 will now indicate a value of R67 275.

Let's use Excel to determine the present value of an ordinary annuity of R7 000 per year for 5 years at an interest rate of 6% per year. We will use the same Excel function but we will use a separate worksheet. Insert R7 000 in Cell B3 as the payment and 6% in Cell B4 as the discount rate. We will place 5, being the number of payments, in Cell B5. The future value is zero and we place this in Cell B6 and place a 0 for type in Cell B7, to indicate that this is an ordinary annuity. This means that the annuity payment is received at the end of each period and starts in this case, in a year's time. In Cell B8, we insert the Excel function, =PV(rate, nper, pmt, FV, type) and refer to the relevant cells where these variables have been placed. The answer is R29 486.55. **Page 2-29**

What is the present value if the annuity was an annuity due? If this is the case, then simply change the value of Type, B7, to 1. The present value indicated in Cell B8 will increase to R31 255.74.

	A	B
1	Present Value of Annuity	
2		
3	Payment	7,000
4	Interest rate	6.00%
5	Number of Payments	5
6	Future Value	0
7	Type	0
8	Present Value	29,486.55
9		
10	What is in Cell B8?	=PV(B4,B5,-B3,-B6,B7)
11		PV(RATE,NPER,PMT,FV,TYPE)

	A	B
1	Present Value of Annuity	
2		
3	Payment	7,000
4	Interest rate	6.00%
5	Number of Payments	5
6	Future Value	0
7	Type	1
8	Present Value	31,255.74
9		
10	What is in Cell B8?	=PV(B4,B5,-B3,-B6,B7)
11		PV(RATE,NPER,PMT,FV,TYPE)

The main functions in Excel to solve time value of money problems are:

Future Value:	= FV (rate, nper, pmt, PV, type)
Present Value:	= PV (rate, nper, pmt, FV, type)
Interest rate:	= RATE (nper, pmt, PV, FV, type)
Number of periods:	= NPER (rate, pmt, PV, FV, type)
Payment:	= PMT (rate, nper, PV, FV, type)

There are many other functions that you can use but we have limited this to the above for now. In most cases you can omit the type or indicate a zero. Excel indicates which variables are important to enter.

In Excel we can also set out the cash flows in Cells with each cell indicating a time period. This is particularly useful for determining the present value of uneven cash flows, but we can use it also for any type of cash flow. We then use the =NPV function which will discount the future cash flows at a specified discount rate. Remember that the NPV function assumes that the first cash flow occurs in one period's time from today.

The NPV function in Excel requires us to indicate the discount rate and the RANGE of cash flows to be discounted and will assume that each cell represents a period. We simply indicate in a Cell =NPV(rate, range). In the following example, the discount rate has been placed in Cell E3 and the cash flows have been placed in Cells C5 to G5. Therefore in Cell B7, you will type =NPV(E3, C5:G5) and the answer will indicate R29 486.55.

	A	B	C	D	E	F	G
1	Present Value						
2							
3	Cash flow	7,000	Interest rate	6.00%			
4		0	1	2	3	4	5
5	Cash flows	7,000	7,000	7,000	7,000	7,000	7,000
6							
7	NPV	29,486.55 ¹					
8		1 =NPV(E3,C5:G5)					
9							
10		0	1	2	3	4	5
11	Cash flows	7,000	7,000	7,000	7,000	7,000	
12							
13	NPV	31,255.74 ²					
14		2 =NPV(E3,C11:F11) + B11					

Page 2-30 If we are discounting a series of cash flows which commence today, then we should indicate the range that should be discounted, in this case C11 to F11 and then add today's cash flow at today's value. In reality we are discounting the cash

flows in periods 1 to 4 by using the NPV function and adding to this the value of Cell B11, which is R7 000.

The NPV function is very useful for discounting a series of uneven cash flows.

	A	B	C	D	E	F	G
17				Interest rate	6.00%		
18		0	1	2	3	4	5
19	Cash flows		4,000	6,000	5,000	3,000	9,000
20							
21	NPV		22,413.26 ³				
22		3 =NPV(E17,C19:G19)					

We will come back to the use of Excel spreadsheets in later chapters. We have just touched the surface in terms of using spreadsheets to solve corporate finance problems.

5 THE ROLE OF INTEREST RATES

When we borrow money, we need to pay for the use of these funds. This is interest and the interest rate is normally indicated as a percentage of the amount borrowed, which is called the principal amount. A company may borrow to invest in plant and equipment and is prepared to pay for the loan because the use of the borrowed funds is expected to generate a value greater than the cost of borrowing. The bank lending the funds to the firm will lend at a rate that is higher than the cost of such funds to the bank. The bank may be borrowing from other banks, its depositors, the South African Reserve Bank or from the issue of bonds. Supply and demand for funds will drive interest rates, although the South African Reserve Bank can influence short-term interest rates by setting the “repo” rate, which is the interest rate that banks borrow from the South African Reserve Bank.

In general, the interest rate, which represents the return required by the lender, is influenced by three main variables. They are:

- **The time value of money.** People prefer to receive money sooner rather than later.

This principle arises because money can be invested to earn more money. Thus the earlier cash is received, the greater is the potential for increasing wealth.

- **The risk of the capital not being repaid.** If there is some uncertainty that the capital amount will be repaid, a premium will be required. When there is a high degree of certainty, for example in the case of loans to the government in the form

of treasury bills, bonds or other “gilt” investments, the premium will be low. This is consistent with the most fundamental principle in financial management: that return must be commensurate with the risk taken.

■ **Inflation.** In times of rising prices it is evident that the purchasing power of money decreases over time. Any lender would expect to be compensated for this decline in purchasing power. If the interest rate did not compensate for inflation the lender would be poorer when the capital is repaid than at the time of the loan.

Liquidity will also impact on the interest rate. If a security is liquid, then it can be converted quickly into cash at a realistic or fair value. Investors will demand a liquidity premium if a company’s bonds are not liquid. The lack of liquidity was an important factor during the global financial crisis.

The rate of interest does not remain constant but depends on expected changes in these variables. [Figure 2.5](#) represents a typical yield curve which reflects the ruling interest rates on investments maturing at various times in the future. The term structure of interest rates, as it is called, reflects the rates of interest charged at any given time for **Page 2-31** borrowing over different periods, from short-term to long-term. It seems reasonable to expect that longer-term loans carry more risk than short-term loans, so we would expect to see an upward-sloping curve. If this is coupled with the expectation that inflation rates are likely to escalate in the future, an upward slope such as that in [Figure 2.5](#) would result.



Figure 2.5 Typical yield curve

So what is the yield curve in South Africa? This is also called the term structure of interest rates. Yields quoted for RSA Government Fixed Coupon bonds on 31 March

2023 were as follows and we can plot a yield curve to depict the term structure of interest rates.

Table 2.10 South African Government Bond Yields

RSA Bond	Mkt Yield	Est. average yrs to maturity	Coupon	Redemption date (s)
SATB6M	8.25%	0.50		30 September 2023
R186	8.42%	3.72	10.50%	21-Dec-25/26/27
R213	10.22%	7.90	7.00%	28-Feb-31
R209	11.07%	13.00	6.25%	31-Mar-3 6
R214	11.40%	17.93	6.50%	28-Feb-41
R2048	11.61%	24.93	8.75%	28-Feb-48

Government bonds pay varying coupon interest rates. A coupon rate of 7% for the R213 issue means that the South African government will pay investors R7.00 per year on a R100 bond. This is normally paid semi-annually, so an investor will receive R3.50 per R100 bond every half year. The coupon rate stays fixed at 7%. However, the price that this bond is trading at means that investors will receive a yield of 10.22% per year. The R213 will be redeemed on 28 February 2031. If we plot these yields, we can see that generally the yield curve on 31 March 2023 is upward sloping. [Page 2-32](#)

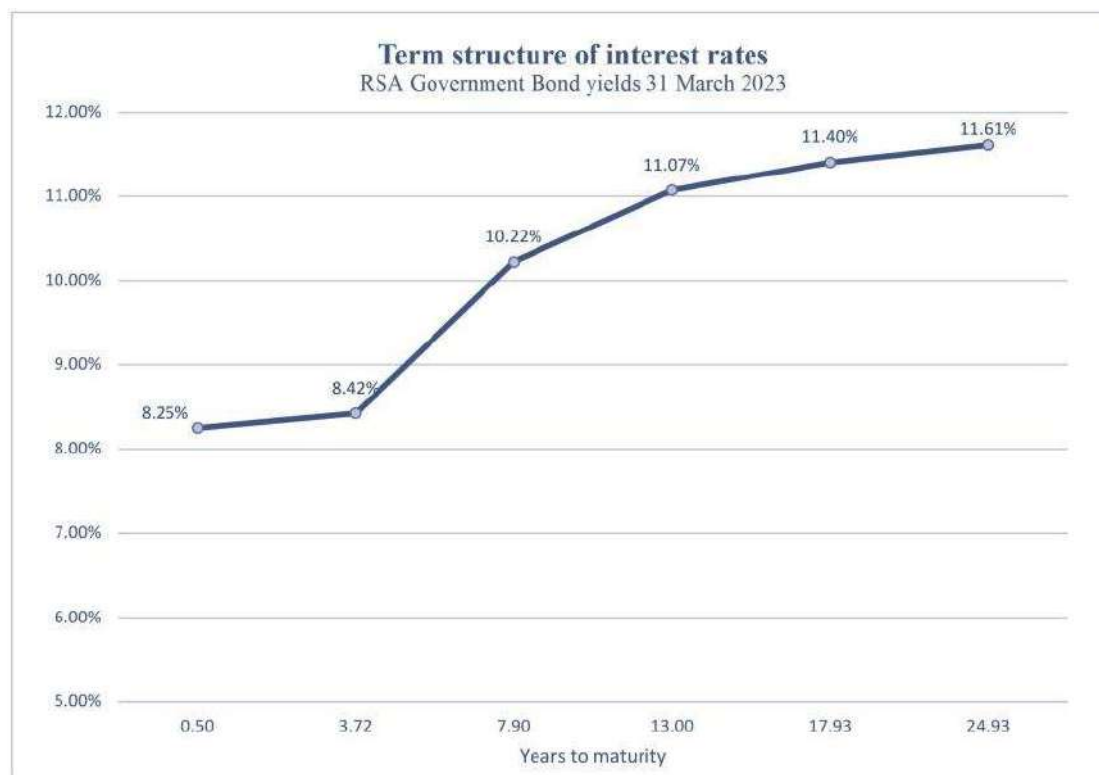


Figure 2.6 . Yield curve in South Africa

A number of theories have been developed to explain the term structure of interest rates. The most commonly quoted theories are the *expectations theory*, the *liquidity preference theory* and the *market segmentation theory*. We can also refer to the precautionary motive for holding liquid or short-term assets.

The expectations theory

Assuming an investor is faced with the choice of investing a sum of money for one year in a bond at a fixed interest rate, or for five years in a five year bond. Which alternative will be chosen? If it is expected that interest rates will decline during the next five years, it will be better to invest for five years as a higher interest rate will be obtained from the investment than if a series of five one-year investments had been made. If this expectation of generally declining interest rates was held by all investors, then the five-year investment would be in greater demand if offered at the same rate as the one-year investment. This demand would force the rate of the five-year investment down, until both alternatives would be equally attractive.

This leads to the question of which factors are likely to contribute to expectations relating to the increase or decrease in interest rates on bonds. The expected inflation rate is a significant factor. An investor would clearly hope at the end of a year to be at least as wealthy as the result of an investment as at the beginning of the year. The interest received should therefore at least compensate for the loss of purchasing power as a result of inflation. In fact, as most interest received is taxable, the investor would expect the *after-tax* interest to at least compensate for inflation. In essence, the expectations theory holds that the *slope of the term structure of interest rates depends on the expected future spot rates of interest*. If it is expected that future rates of interest will be higher, the yield curve will be upward-sloping.

The liquidity preference theory

A second theory which attempts to explain the term structure of interest rates introduces an element of risk. When dealing with investments such as government bonds, the risk is not Page 2-33 related to default on the capital invested, but rather to interest rate fluctuations. For example, assume you hold both a 6-year RSA Government and a one-year RSA Government Bond, each with a face value of R100 and each with a coupon rate of 8%, when the market rate of interest is also 8%. If the market interest rate rises, the negative effect on the value of the 6-year RSA Bond would be far greater than on the one-year RSA Bond. The interest rate risk is greater,

the longer the period to maturity. [Figure 2.7](#) illustrates the effect of changes to the market interest rate on the price of two 8% RSA Bonds with differing maturity dates.

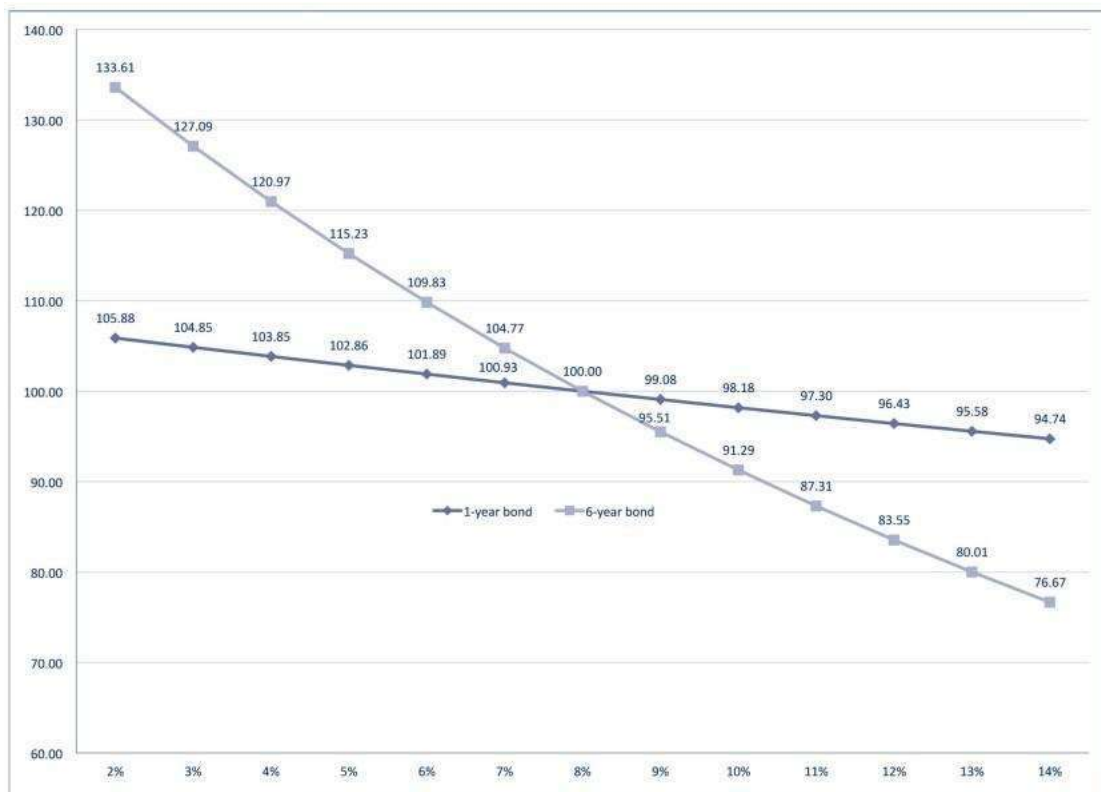


Figure 2.7 Value of 8.0% RSA Bonds on 1 October 20x6

As these investments are RSA Bonds, the risk of default on repayment is virtually zero. However, there is considerable risk for the longer-term bond if market interest rates increase. If the market rate on 1 October 20x6 is 8%, the value of both bonds will be the same at R100. If the market interest rate rose on that date to 11%, for example, the value of the 6 year Bond would decline by R12.69(100.00 – 87.31) considerably more than the decline in value of the 1 year Bond due at the end of September 20x7, of only R2.70 (100.00 – 97.30).

The reasons for the greater variability of the 6 year bond stems from the difference in the timing of the cash flow between the two investments. Investors holding the 6 year bond are locked into receiving a 8% return on R100 for 5 years longer than holders of the 1 year bond.

As investors prefer certainty to uncertainty, the theory holds that investors will generally prefer short-term investments to long-term ones. As a result the short-term rates will tend to be lower than long-term rates because of the higher demand for short-term investments. However, short-term bonds are exposed to reinvestment rate

risk. When we need to reinvest maturing short-term bonds (“rolling-over”) then we are uncertain what the interest rate will be at the time. Page 2-34

The market segmentation theory

The market segmentation theory is based on the premise that different investors have differing investment preferences as to timing due to legal, regulatory, business and personal motives. Life insurance companies and pension funds have long term liabilities and prefer to invest in securities which have distant maturity dates. Banks may invest in short-term securities as most of their liabilities are of a short-term nature. This theory implies that interest rates are determined by demand and supply factors in these market segments. The other factor is that the Reserve Bank will have a greater influence over short-term rates rather than long term rates.

6 APPLYING THE TIME VALUE OF MONEY PRINCIPLES TO BONDS

The principles of the time value of money are applied in the valuation of bonds. A bond is a financial instrument issued by government and companies to raise funds. The bond or debenture will stipulate that the issuer is obliged to pay the bond holder a fixed interest or coupon rate until the maturity of the bond when the capital amount, usually R100, will be repaid to the bond holder. The bond document is a negotiable instrument, meaning that it can be traded during its life span by investors wishing to buy or sell bonds. The name dates back to earlier days when the bond certificate had attached to it, a series of coupons, which entitled the bondholder to collect the interest by surrendering each coupon on the due date.

A Treasury bond is in fact a loan to the government for a number of years. The name of the bond is the RSA Government bond; the annual interest rate payable may be 7.5% (the coupon rate); the year in which the capital amount of the loan will be repaid may be 2036 (the year of maturity). An investment in a bond is thus the purchase of a stream of expected cash flows. The cash flows are virtually certain, given the credibility of the issuer of the bond, and are thus considered to be a low risk investment. Investment in bonds are most typically made by large pension funds, to ensure that the fund has an allocation of low risk investments with a certain interest cash flow to the fund.

The terminology and valuation procedures relating to bonds and bond portfolios are complex and detailed coverage is not undertaken here. We will focus on applying the

time value of money principles we have studied so far.

- A bond is a financial instrument offering two distinct cash flows. Firstly, it offers a stream of regular cash flows in the form of coupon payments. Using time value principles, this is identical to annuity cash flows on regular, equally spaced time periods in the future. Secondly the repayment of the nominal value will occur on maturity date. Using time value principles, this is the equivalent of a single sum payment at a determined future date.
 - An investor who purchases the bond and holds the bond to maturity has a high degree of certainty that each coupon amount of R3.75 will be paid twice a year (semi-annually) on the specified dates, and that the face value or par value will be returned on maturity date. The investor will thus receive a coupon return of 7.5% per year.
 - Bonds may be bought and sold during the period from issue to maturity. It is not uncommon for a bondholder to sell a bond to a willing buyer, who wishes to hold bonds at a time when a new issue is not available.
 - The selection of the coupon rate is dependent upon economic factors at the time of the issue. Prevailing interest rates, inflationary expectations and the expected return of investors from bonds in a similar risk category are the most important factors. These determine the required rate of return of investors in bonds. The required rate of return is also referred to as the Yield to Maturity (YTM).
- Page 2-
- 35
- Even more significant in understanding what follows, is the awareness that interest rates in the economy may change during the life of the bond. Such changes will impact on the YTM, and therefore on the value of the bond. Note, however, that if the bondholder simply retains possession of the bond, the return to the bondholder over the life of the bond will be 7.5% per year.
 - There are consequently three key variables that impact on the value of a bond. They are firstly the coupon rate at which the bond was issued, secondly the ruling market rate in the economy at the time of valuation, which is reflected in the required YTM, and thirdly the period of time remaining to maturity. Other variables include a firm's credit risk and the bond's rating.

We are now in a position to explore the valuation of bonds. We will value an RSA Government security which has a fixed coupon rate of 10% per year, and a face value of R100. Coupon interest is payable semi-annually and the maturity date is the end of October 2028. Assume the current date is 1 November 2024. The current market yield (yield to maturity) on similar securities is 8% per year (4% per half-year)². What is the value of this RSA Government security? The future cash flows from investing in this security can be depicted as follows:

	A	B	C	D	E	F	G	H	I
7	Treasury Fixed Coupon	Apr-25	Oct-25	Apr-26	Oct-26	Apr-27	Oct-27	Apr-28	Oct-28
8	10%, Oct 2028								
9	Cash flow from Coupon	5	5	5	5	5	5	5	5
10	Cash flow at Maturity								100
11		5	5	5	5	5	5	5	105

The discount rate to apply to the future cash flows will be 4% per half-year (8%/2). We will assume that the cash flows will occur at the end of April and October of each year.

The valuation of the bond can be done in two steps using the principles of present value calculation. Step 1 entails discounting the coupon cash flows, which are an annuity, using Formula 2.10. Step 2 entails discounting a single sum, the nominal value to be received on maturity date, using Formula 2.3.

Step 1: Present value of an annuity of R5 for 8 periods at 4% per half-year.

$$PVA = PMT \times \left[\frac{1 - \frac{1}{(1.04)^8}}{0.04} \right]$$

$$\begin{aligned} PVA &= R5 \times 6.7327 \\ &= R33.66 \end{aligned}$$

Step 2: Present value of a single amount of R100 to be received in 8 periods' time at 4% per half-year.

$$\begin{aligned} PV &= R100 \times 0.7307 \\ &= R73.07 \end{aligned}$$

$$\begin{aligned} \text{Sum of two streams of cash flows} &= R33.66 + R73.07 \\ &= R106.73 \end{aligned}$$

We can also use the present value factors for each cash flow and then sum the present values to obtain the value of the bond. Otherwise, use the NPV function in Excel to

discount the future cash flows at 4% per period.

	A	B	C	D	E	F	G	H	I
7	Treasury Fixed Coupon	<i>Apr-25</i>	<i>Oct-25</i>	<i>Apr-26</i>	<i>Oct-26</i>	<i>Apr-27</i>	<i>Oct-27</i>	<i>Apr-28</i>	<i>Oct-28</i>
8	10%, Oct 2028								
9	Cash flow from Coupon	5	5	5	5	5	5	5	5
10	Cash flow at Maturity								100
11		5	5	5	5	5	5	5	105
12									
13	PV factor $1/(1+r)^n$	0.9615	0.9246	0.8890	0.8548	0.8219	0.7903	0.7599	0.7307
14	Present Value	4.8077	4.6228	4.4450	4.2740	4.1096	3.9516	3.7996	76.7225
15	Value (sum of PVs)	106.73							
16									
17	Net Present Value (Excel)	106.73	[Using the =NPV function =NPV(0.04,B11..I11)]						

Page 2-36 The value of the bond is trading at a higher value than par because investors are requiring a return of 8% per year (4% per half-year) and the bond is offering a coupon rate of 10% per year (5% per half-year). A price of R106.73 means that investors will receive a return of 8% per year (4% each half-year) until maturity. If market interest rates rise to 12%, then the value of the bond would fall to R93.79, which is below par as the bond is offering only 10% per year whilst the market is offering at 12%. At a price of R93.79, investors will receive a return of 12% per year (6% per half-year). If the market interest rate is 10% per year, then the market value should equal the par value of the bond. If we redo the valuation at 10% (5% per half-year) we will see that this is true.

	A	B	C	D	E	F	G	H	I
7	Treasury Fixed Coupon	<i>Apr-25</i>	<i>Oct-25</i>	<i>Apr-26</i>	<i>Oct-26</i>	<i>Apr-27</i>	<i>Oct-27</i>	<i>Apr-28</i>	<i>Oct-28</i>
8	10%, Oct 2028								
9	Cash flow from Coupon	5	5	5	5	5	5	5	5
10	Cash flow at Maturity								100
11		5	5	5	5	5	5	5	105
12									
13	PV factor $1/(1+r)^n$	0.9524	0.9070	0.8638	0.8227	0.7835	0.7462	0.7107	0.6768
14	Present Value	4.7619	4.5351	4.3192	4.1135	3.9176	3.7311	3.5534	71.0681
15	Value (sum of PVs)	100.00							
16									
17	Net Present Value (Excel)	100.00	[Using the =NPV function =NPV(0.05,B11..I11)]						

A useful point to note here is that investors and bond portfolio managers aim to forecast the timing and magnitude of interest rate changes in the economy. It is evident that if a fall in market interest rates is correctly forecast, the purchase of bonds prior to that date will result in a significant capital gain on sale, after the YTM responds to the rate fall.

We saw in a previous example that the price volatility increases with the term to maturity. Therefore the value of a 5-year bond will change to a greater extent than a 1-year bond. However, the price volatility is also linked to the coupon rate. The lower the coupon rate the more volatile will be the changes in price to changes in interest

rates. This simply reflects the fact that you can reinvest the coupon payments at the market rate. A composite measure of term to maturity and coupon is called a bond's duration and the greater the duration of a bond, the more sensitive a bond's value will be to changes in market interest rates.

The following principles summarise the use of time value principles and the impact on bonds:

- A bond offers two cash flows – a series of coupon payments at a specified constant rate, and the repayment of the nominal value of the bond at maturity.
- When the yield to maturity (YTM) differs from the coupon rate, cash flows are discounted at the YTM in order to establish the present value of the bond.
- If the YTM moves above the bond coupon rate, the value of the bond falls below its par value. If the YTM moves below the coupon rate, the value of the bond rises above its par value. Page 2-37
- When the YTM differs from the coupon rate, the length of time remaining to maturity of the bond impacts on the magnitude of the difference between the nominal value and the market value.
- The longer the term to maturity, the greater the impact on present value. This was graphically presented in [Figure 2.7](#).
- Although not illustrated here, it also follows that if two bonds have different coupon rates, the value of the bond with the lower coupon rate will change relatively more than the bond with the higher coupon rate when the market YTM rises or falls.
- Finally, apart from the typical bond as described, bonds may come with other characteristics. Bonds could have call options, which applies to some corporate bonds. This allows the company to redeem the bonds at the option of the company, prior to maturity date. RSA Government securities will pay coupons semi-annually (twice each year). The cash flow line is thus constructed in periods of 6 months rather than annually. Another example is a zero coupon bond, where there are no coupons. Only the future par value, repayable on redemption date is therefore discounted to the present by the YTM. We will now analyse bonds without a redemption date.

Perpetuals and 100-year bonds

A company (or country) may issue perpetual bonds which do not have a maturity date and there is no obligation to redeem the face value of the bond. The company is required to make the coupon interest payment forever. As there is no redemption of principal these bonds are seen as more akin to equity. It is useful for banks to issue these types of bonds to bolster their capital in terms of *Basel III*. Sukuk issues (a type of perpetual instrument) are popular in Islamic countries and form a key component of Islamic finance as there is no redemption of principal and therefore Sukuk issues may be viewed as equity. A perpetual bond's value is simply:

$$\text{Value} = \text{Coupon payment} / \text{market yield}$$

The value of a perpetual bond will be highly sensitive to changes in interest rates. If we use the same example set out in [Figure 2.7](#) (coupon 8%, principal R100), a rate rise from 8% to 11% resulted in a fall in the market value of the one-year bond from R100 down to R97.30 whilst the 6-year bond saw a fall from R100 down to R87.31. If this was a perpetual bond, then a 3% increase in the market yield would see a fall in value from R100 to R72.73 [R8/0.11]

Market Yield	Value 1yr bond	Value 6yr bond	Perpetual
5%	102.86	115.23	160.00
8%	100.00	100.00	100.00
11%	97.30	87.31	72.73

However, for investors expecting interest rates to fall, then the investment in perpetuals can offer significantly higher returns. If the interest rate falls from 8% to 5%, then the increase in value will be 60% for the perpetual bond, which is higher than the 15% increase for the 6-year bond value and the 2.86% increase in the value of the 1-year bond.

Companies and countries have also started to issue 100-year bonds which are also highly sensitive to interest rate changes but do have a redemption date. Just do not expect to get your money back for you will probably not be around.

The advantages of perpetuals are that these bonds tend to offer a higher yield. However, it is important to note that a number of perpetuals are callable by the company issuing the bonds. This means that the company will have the right but not the obligation to redeem a “perpetual” after a fixed period of time. Page 2-38

Let's come back to our example of an 8% perpetual bond but we will now assume that the company can call/redeem the bond issue after five years. The company has the right to pay you back but you cannot demand repayment of principal and simply have to sell the perpetuals if you wish to recover the market value of the bond. If interest rates fell from 8% to 5%, then it is highly likely that the company will call/redeem the bonds in five years' time at par and reissue new bonds at the lower rate of 5%. All this means is that when you value the bond, assume that the higher coupon rate will only last until the call date if rates have fallen. If this happens then this means that you are in effect valuing a redeemable bond with a likely redemption date which is the call date.

If interest rates rise above the coupon rate, then the company will not call or redeem and you can expect that you will receive the coupon payment forever which means that you will not be able to reinvest your principal at the higher market rate. So, investors have downside and limited upside as the company will call and refinance the bonds if market interest rates fall. In exchange, the interest rate will be higher at least until the call date. Just do not expect to receive the interest yield beyond the call date if market interest rates have fallen.

FROM THE REAL WORLD

Nedbank issued R1bn perpetual (callable) subordinated notes (NGT111) on 22 April 2022 at a floating (variable) rate of JIBAR plus a margin of 3.79%. The early redemption (call) date is 23 April 2027 or any interest date falling after the call date. The JIBAR rate on 14 April 2023 was 7.96% which would result in a total interest rate payable of 11.75% to investors in these notes. If Nedbank had issued fixed rate notes, and if interest rates fell and/or market credit spreads (the margin) fell, then Nedbank would be expected to call in the fixed rate notes on the call date and reissue notes at the lower rate and/or lower credit spread. For the NGT111 issue, these notes are floating rate notes and so the main motive for calling and reissuing the notes would be if Nedbank can obtain lower credit spreads in the future.

JIBAR(Johannesburg Interbank Average Rate) is a benchmark overnight rate which banks are prepared or expected to lend and borrow from each other. JIBAR will be replaced by the overnight rate, ZARONIA (South African Rand Overnight Index Average), which is based on actual lending and borrowing transactions rather than JIBAR which is a forward looking rate.

Example 2.18: Perpetual bonds, yield to call and yield to maturity

MTX issued perpetual bonds on 31 December 20x0 with a face value of R100 at a coupon interest rate of 10%, payable in arrears, which are callable after 5 years at par. At the time of issue, the market rate for this type of bond was 10% but one year later

on 31 December 20x1 (after the coupon payment for 20x1), interest rates have fallen to 6% and this bond is now trading at a market value of R110.

1. What would you expect the value of the bond to be on 31 December 20x1 (after the coupon interest payment) if MTX was not permitted to redeem after 5 years?
2. What is the likely yield of these bonds at 31 December 20x1 (after the coupon interest payment) if we consider that the company is able to redeem (call) the bonds in 20x5?
3. Assume the same information except that the bonds are not perpetual but are 10-year bonds and are not callable. What would the Yield to Maturity be if the price of the bond was trading at R110 on 31 December 20x1 (after the coupon interest payment)?

Solution

1. The value at 31 December 20x1 would be expected to be R166.67 ($R10/0.06$) as we are effectively valuing a perpetuity. **Page 2-39**
2. It is likely that the company will call (redeem) the bond. The reason for the increase in price, is due to the fall in interest rates and the company will be able to reissue bonds at a lower interest rate at the end of 20x5 if interest rates are expected to remain low. The yield to the call date and the yield an investor will earn is expected to be 7.04%.

	A	B	C	D	E	F	G
17	Yield to Call	20x1	20x2	20x3	20x4	20x5	
18	Price of bond	-110					
19	Cash flow from Coupon		10	10	10	10	
20	Cash flow at Maturity					100	
21		-110	10	10	10	110	
22							
23	Yield to Call	7.04%	[Using the =IRR function in EXCEL, =IRR(B21..F21, 0.10)]				

3. If the investor pays R110 for the bond and the maturity date is in 9 years' time, then the yield to maturity will 8.37%.

	A	B	C	D	E	F	G	H	I	J	K
9	Yield to Maturity	20x1	20x2	20x3	20x4	20x5	20x6	20x7	20x8	20x9	20x10
10	Price of bond	-110									
11	Cash flow from Coupon		10	10	10	10	10	10	10	10	10
12	Cash flow at Maturity										100
13		-110	10	10	10	10	10	10	10	10	110
14											
15	Yield to Maturity	8.37%	[Use the =IRR function in EXCEL, =IRR(B13..K13, 0.09)] [0.09 is an estimated rate]								

SUMMARY

In this chapter, we have focused on an important and fundamental concept in financial management, namely the fact that money has a time value. The time value of money is one of the variables that affects the interest rate. Two other variables that affect the

interest rate are expected inflation and risk. The risk attached to an investment relates to the expected stream of cash flows from the investment as well as the ultimate return of the capital sum invested.

If money is being loaned or invested, the investor expects the future value to be greater, because a reward for permitting another person the use of the funds must be received, in addition to the repayment of the original amount invested. The future value is determined by compounding the interest at the agreed rate. It may be important to calculate the effective annual rate when comparing different options. This differs from the quoted rate when interest is compounded more frequently than once a year. It is the effective rate which is relevant for purposes of comparison and evaluation. Two basic types of future values can be identified. They are the future value of a single amount investment and the future value of an annuity.

The determination of present values is even more important in financial management than future values. Present values of cash flows will clearly be lower if the cash flows are to be received in the distant future. The interest factor is therefore used to discount expected future cash flows to their present value. Some present value calculations require special attention. They are the cases of annuities due, deferred annuities, uneven streams, perpetuities, and instalment loans.

Mortgage loan amortisations and retirement planning are two cases where time value of money principles are applied. The valuation of bonds illustrates another significant area for applying these principles. A bond is essentially a long-term loan, which offers the holder a series of stable cash flows during its life and a repayment of the nominal amount on maturity date. It was shown that the value of a bond may fluctuate dependent upon the coupon rate on the bond, the yield to maturity required by investors and the period of time remaining to maturity.

We used examples to illustrate the more important formulae and to demonstrate the use of the tables. We also expanded on the use of financial calculators and Excel Spreadsheets **Page 2-40** to solve time value of money problems. The concepts developed are not complex. However, great care must be taken to formulate each problem correctly and to recognise the type of investment which a particular situation reflects. Many of the concepts developed in this chapter will be used extensively later in the text.

SELF-STUDY PROBLEMS

S2.1

You deposit R10 000 in a bank account which is paying 4.8% per year, interest compounded annually. How much will you have accumulated in the account in 5 years' time? What will you have accumulated if interest is compounded monthly?

S2.2

You have purchased a motor car for R120 000 and you have obtained a car loan for the total amount, which requires you to pay this amount over 5 years at an interest rate of 7.2%. If interest is compounded monthly, determine the monthly payment required over the 5 years if the first payment is due immediately.

S2.3

What is the present value of a zero coupon bond, with a par value of R100, which is due to be redeemed in 10 years' time, when the current market interest rate for such bonds is 6%, interest compounded semi-annually?

S2.4

You wish to purchase an apartment which is situated in a tree-lined avenue. The purchase price, with costs, is R710 000 and you are able to obtain a 100% mortgage loan at an interest rate of 6%, interest compounded monthly. The term of the loan is 20 years. Assume that property values are expected to rise at a rate of 9% per year (0.75% per month). You will be able to rent out the apartment after costs at a rate of R4 000 per month for the first year. Interest and rent are payable at the beginning of each month.

Required:

What is the expected value of the apartment in 20 years' time? What is the mortgage loan repayment at the beginning of each month? What is the net amount you have to pay in each month?

Solutions to Self-study Problems

S2.1

The future value at an interest rate of 4.8% with annual compounding is:

$$FV = 10\,000 (1.048)^5 = R12\,641.73$$

The future value with monthly compounding is determined by using a monthly rate of 0.4% (4.8%/12) and 60 months (5 × 12):

$$FV = 10\,000 (1.004)^{60} = R12\,706.41$$

The higher amount is due to interest earning on interest more often over the period. **Page 2-41**

S2.2

This is an annuity due. The monthly payment on the car loan is determined by using 60 months and an interest rate of 0.6% (7.2%/12). Remember with an annuity due there is one less discounting period.

$$120\,000 = PMT \times \left\{ \left[\frac{1 - \frac{1}{(1.006)^{59}}}{0.006} \right] + 1 \right\}$$

$$120\,000 = PMT [50.5637]$$

$$PMT = 120\,000/50.5637$$

$$PMT = R2\,373.24 \text{ per month}$$

If we use a financial calculator, then press 12 followed by SHIFT and P/YR to change the financial calculator to monthly compounding. Also, press the SHIFT key and the BEG/ END key to change the mode to Beginning of the period, i.e. to indicate to the financial calculator that the cash flows will occur at the beginning of each period.

N	I/YR	PV	PMT	FV
60	7.2	120 000	-2 373.24	0

The payment that is required to repay the loan and interest will amount to R2 373.24 per month at the beginning of each month for 5 years.

S2.3

A zero coupon bond is a bond with no coupon payments and the value is dependent on the redemption date and the par value. The par value is R100 and the redemption date is in 10 years' time. The discount rate is 3% per half year and there are 20 six-monthly periods. The value of the zero coupon bond is determined as follows:

$$PV = FV/(1.03)^{20}$$

$$PV = 100/1.8061$$

$$PV = R55.37$$

S2.4

The expected value of the apartment in 20 years' time is:

$$FV = 710\,000 (1.0075)^{240}$$

$$FV = 710\,000 (6.009152)$$

$$FV = R4\,266\,498$$

If property keeps on appreciating at a rate of 9% per year, then the value in 20 years' time is expected to be over R4.2m. If the rate of appreciation is 6% per year then the value in 20 years' time will be R2.35m, so 3% makes quite a difference.

The monthly repayment due at the beginning of each month is determined by applying the annuity due formula:

$$710\,000 = PMT \times \left\{ \left[\frac{1 - \frac{1}{(1.005)^{239}}}{0.005} \right] + 1 \right\}$$

$$710\,000 = PMT/(140.2787)$$

$$PMT = 710\,000/140.2787$$

$$PMT = R5\,061.35$$

Page 2-42 You will need to pay in R1 061.35 each month after taking into account the net rental of R4 000 per month. At a marginal tax rate of 40%, the interest and other costs would be deductible for tax purposes. Remember that the rental should also grow over time.

Using a calculator and ensuring that you use a beginning of period mode and monthly compounding as in S2.2 above, the payment is determined to be R5 061.35.

N	I/YR	PV	PMT	FV
240	6	710000	-5061.35	0

APPENDIX 2.1: RETIREMENT PLANNING: LIVING ANNUITIES, THE 4% RULE AND THE LOOMING PENSION CRISIS

I advise you to go on living solely to enrage those who are paying your annuities. It is the only pleasure I have left. Voltaire. Money is something you got to make in case you don't die.

Max Asnas. You can be young without money but you can't be old without it. Tennessee Williams.

Time value of money principles are crucial in structuring retirement pensions and annuities. We will expand on our example in the chapter by explaining the options facing South Africans considering retirement. Persons approaching retirement will have contributed to a retirement fund over their working lives and will then need to ensure that any accumulated sum generates an adequate income to provide for them in their years of retirement. A retired person will have two main alternatives, to either select a *guaranteed life annuity* or select a *living annuity*³.

A **guaranteed life annuity** is an insurance product whereby an insurance company will guarantee a retiree a set annuity income for the life of the retiree. This may include an adjustment for inflation, or fixed escalations and may offer other options. A person may have accumulated within a retirement fund a total amount of say R3.0m. The retiree transfers this amount to an insurance company who may promise, for example, to pay a pension annuity of R14 000 per month increasing by inflation each year for the life of the retiree. If the retiree dies after 2 years, then the capital normally dies with the retiree. However, the insurance company will lose if the retiree lives for much longer than expected in terms of the actuarial life expectancy tables. The advantage is that the pension annuity will last for life and is guaranteed (as long as the insurance company remains in business). The insurance company will pool its longevity risk across many annuitants. The quote above by Voltaire relates to a guaranteed life annuity. However, a guaranteed life policy has the following disadvantages:

- Loss of liquidity and control over one's assets. Once there is a transfer of the retirement sum, a retiree can no longer recover the amount or have access to it. There may be unexpected medical costs or other expenditure and there is no flexibility to access the funds or pay for unexpected emergency expenditure.
- There is no legacy and a retiree will be unable to leave money to his or her heirs.
- High costs and lack of transparency.
- Insurance companies tend to focus on investing in fixed-income securities to support a lifetime pension and this will normally offer lower returns (at a lower risk) than other investment alternatives such as unit trusts which include ordinary equities. However, insurance companies are starting to insert equity options and perhaps providing an income component which is variable depending on the performance of the underlying equity assets.
- Although unlikely, insurance companies sometimes fail. Annuities are paid over many years and therefore the question of whether the company will be around in 25 years' time

is relevant.

Page 2-43 It may be possible to plan around these limitations by taking out an initial part of the retirement sum as a tax-free amount and investing this outside the pension/annuity structure. Also, there is greater flexibility these days with guaranteed life annuities and options may include a guaranteed period or provide an annuity over the lives of both spouses. However, the impact of such options on the promised annuity can be significant. The main benefit of a guaranteed annuity is that it does provide protection against outliving one's assets.

The alternative and significantly more popular retirement option in South Africa is a **living annuity**. This is more aligned to an investment product but is subject to regulatory limits set by government. A living annuity is offered by financial institutions and asset management firms whereby a retired person will withdraw a regular income but this annuity amount is not guaranteed as it depends on the lifespan of the retiree and the performance of the underlying investments. This means that the retiree will bear the risks of longevity, investment returns and inflation. In terms of the Income Tax Act which sets out the definition of a living annuity, a retiree will be able to withdraw between 2.5% and 17.5% of capital per year. For example, the following is from Allan Gray's Living Annuity terms and conditions (effective 9 March 2018, version 12.2)

Your annuity income may differ from year to year. Neither the amount of the annuity income, nor the return of the underlying investment, is guaranteed. The market value of the investment account may fluctuate and you carry the risk of your capital and income being reduced.

A higher initial capital amount and a low drawdown per year will improve the chances that the fund will provide a pension for life but for many pensioners the reality is that they may run out of money during their lifetimes due to high drawdowns, low returns or living longer than expected. A rough guide that has been often promoted is that the initial drawdown should be 4% for the first year, and this initial amount can thereafter increase by inflation to retain its purchasing power. A living annuity provides flexibility and if you die early, then there will be money to leave to your family. Investment returns and capital gains in the fund are tax free but pension payments are taxable. Let's refer to an example and we will use annual amounts and include inflation. We will ignore for now the regulatory constraints that the pension should be between 2.5% and 17.5% of the value of the investments in any year but we will come back to this later.

Let's assume that Lerato Khumalo, a Chartered Accountant, is coming to the end of her working life and has accumulated a sum in her defined contribution pension plan of R2.5m and is looking forward to retirement. She needs a pension with a current value of R228 000 per year to live on and this annual pension needs to increase at the expected inflation rate of 5.0% per year. Assume the annuity is payable at the end of each year. She will invest in a diversified bond and equity portfolio which is expected to generate a return of 10.0% per

year⁴. She is 65 years old and estimates that she will need a pension for 30 years. She owns her own home and there is no mortgage loan, so if she lives longer than 30 years, then she will sell her home. Planning for such a long retirement period means that she should be able to leave a legacy to her heirs as well.

If we apply Formula 2.13 for a growing annuity, we can determine that the capital amount required to enable her to acquire an expected annuity with a current value of R228 000 per year (or R239 400 at end of the first year) is:

$$PV = 228\,000 (1.05) \times \left[\frac{1 - (1.05)^{30}/(1.10)^{30}}{0.10 - 0.05} \right]$$

$$PV = R3\,602\,088$$

Page 2-44 This is important as it indicates that a sum of R2.5m is not sufficient to provide the necessary income over her remaining life. It is important to know this upfront. If we redo the above calculation with an annuity (with today's value) of R158 241.56 per year increasing thereafter by inflation so that the annuity at the end of the first year is R166 153.64, then the required initial amount would be R2.5m. Is it possible for Lerato to obtain part-time work after retirement to make up the annual difference between R166 153.64 and R239 400.00 ($R228\,000 \times 1.05$)? Is it possible to downsize and move into a smaller property and perhaps invest the difference to ensure a total income of R239 400 per year from year one and which will grow by inflation thereafter?

Let's now change our assumptions and we will now determine that Lerato has been able to accumulate exactly R3 602 088 at retirement date. In this case, the value of her retirement assets at the **end** of each year is set out in [Figure 2.8](#). For example, at the end of years 1 and 2, the value of the assets is equal to the beginning asset value multiplied by one plus the return of 10% less the annuity for the first and second years:

$$R3\,602\,088 \times 1.10 - R239\,400 = R3\,722\,897$$

$$R3\,722\,897 \times 1.10 - R251\,370 = R3\,843\,816$$

What is interesting is that the value of Lerato's investments will continue to grow until year 16 where it will reach R5 002 369m and even by year 24, the nominal value of her investments at R3 760 860 will still be very close to what she began with, but will fall swiftly in value in the last few years in order to fund the growing annuity. In the first 16 years, the return on the investments will be higher than the inflation-adjusted annuity and this will increase the value of the investments. However, from year 17, Lerato's annuity will exceed the income from the investments and she will begin to draw down capital. Therefore, there will be a growing capital contribution each year to the annuity, due to lower income and a rising annuity payment, until the value of the investments is zero at the end of year 30.



Figure 2.8 Value of retirement assets during retirement period

In [Figure 2.9](#), we present the growing annuity, and the income per year as well as the capital contribution. Until year 16, the capital contribution to the annuity is negative, which means that we are accumulating capital (as income is greater than the annuity) but from year 17, we see a growing capital drawdown to make up the required annuity amount. For example, in year 8, investment income is R442 619, whilst the pension annuity is R336 860, which results in savings of R105 759 which will increase the value of the retirement assets by the same amount.

Page 2-45 In year 18, the annuity is R548 709 and income is R498 003, which means that the balance of R50 707 is drawn from capital and the value of the investment assets will fall by the same amount. In year 30, only R89 582 is from income and the balance of R895 821 (to make up the required annuity of R985 403) is from capital, which will reduce the value of the retirement assets to zero.

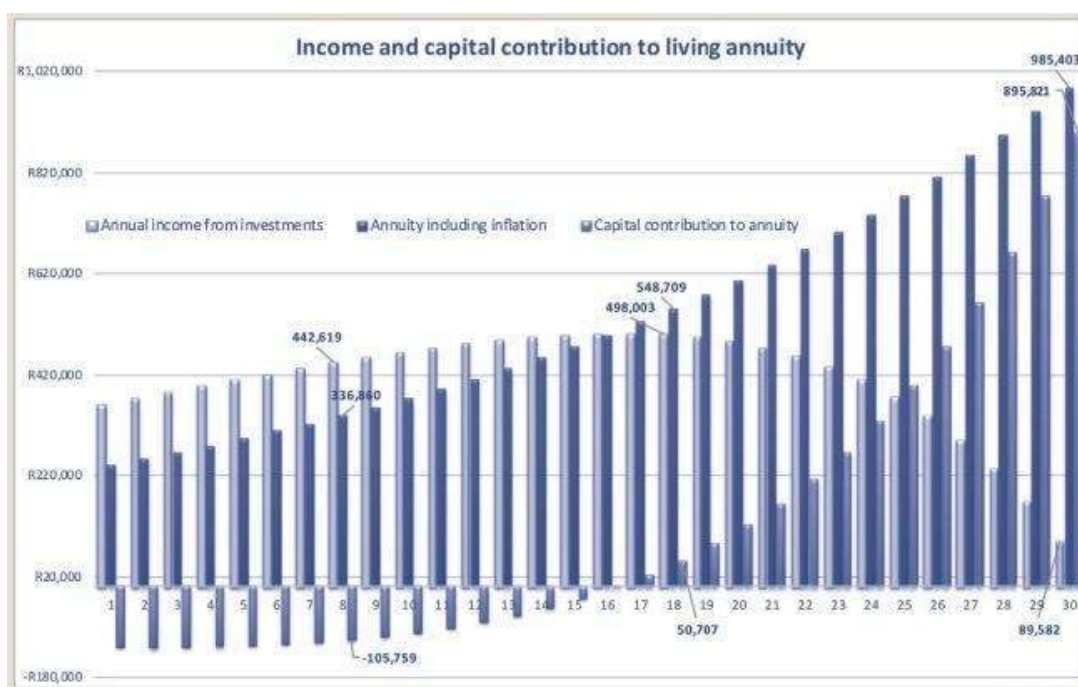


Figure 2.9 Living annuity with annual returns on retirement assets and capital contributions

If Lerato is able to accumulate R3 602 088 by retirement date, then she will have funded her retirement perfectly. However, this is based on some perfect assumptions. If she dies early, then any remaining balance in the fund will be left to her heirs, with the house. If she lives longer, then this may create challenges for her, but she has a plan B which is to sell her house and move into rented premises or enter into a reverse mortgage whereby she borrows against the house and repayment only occurs on her passing away or on the sale of her house. Remember, she is planning for this upfront so that it does not become an emotional issue for her later on. There are other options such as renting out a room or renting out the house and moving into a smaller apartment at a lower rental. These options should be explored a number of years before the 30th year of retirement.

There are other assumptions. The market has returned about 10% per year but this may not apply in the future. If market returns are lower and the inflation rate is higher, then this may place Lerato in a difficult position and so there should be some slack built in or Lerato should try and save part of the annuity or perhaps reduce the required annuity in order to ensure that there is a margin of safety. Currently, her home performs this role. We have assumed a return of 10% after fees. What happens if the 10% return is before fees? A financial institution may charge around 2% per year and this can have a material impact on the value of the retirement assets over time. Only a few financial institutions or asset management firms are able to beat the market index over time *after costs*. Investing in a low-cost market index may therefore be a good alternative unless the manager can show outperformance over a sustained period of time. Page 2-46

The 4% rule and the sustainability of a living annuity

A critical question facing any retiree is how much to withdraw from the fund each year. If a retiree withdraws too much, then a retiree will run out of funds and may end his or her later years in poverty. If the withdrawal rate is too low, then a retiree may experience hardship or lower living standards during retirement. How much can a retiree safely withdraw each year in real terms so that he or she does not run out of money? The annual withdrawals are affected by equity and bond returns, asset allocation, inflation rates and the expected lifespan of a retiree.

William Bengen, a USA financial planner, decided to study historical equity returns, bond returns and inflation rates over a long period for portfolios created each year since 1926 to test initial withdrawal rates over a retirement period of 30 years. The initial withdrawal amount was then increased by the inflation rate each year to determine the annual withdrawal amount for each year. The initial capital value was increased by the actual returns from a portfolio of 50% bonds and 50% equities, which was rebalanced each year, and the value of this portfolio was reduced by the withdrawal amount each year. Bengen writes that

“assuming a minimum requirement of 30 years of portfolio longevity, a first-year withdrawal of 4%, followed by inflation-adjusted withdrawals in subsequent years should be safe”². This article led to the 4% rule that financial planners tend to use around the world to determine a safe withdrawal rate. Bengen also indicated that a portfolio of 75% equities would improve the longevity of a portfolio if investors were able to cope with a higher level of volatility. So, what about South Africa?

Luckily for us, Michael Summerton, formerly of Allan Gray, did a similar study for South Africa and considered equity returns, bond returns and inflation rates for South Africa⁶. Summerton, in the Allan Gray study, went back to 1900 and created 84 30-year retirement periods to take us up to 2014. [Table 2.11](#) presents the results from Bengen’s study and from the Allan Gray study.

Table 2.11 Withdrawal rates and the number of years a portfolio will last

Bengen				Allan Gray			
Withdrawal rate (initial)	3%	4%	5%	Withdrawal rate (initial)	4%	5%	4%
Equities in portfolio	50%	50%	50%	Equities in portfolio	55%	55%	55%
Increase in withdrawal	inflation	inflation	inflation	Increase in withdrawal	inflation	inflation	above inflation
Number of years				Number of years			
30 or more	100.0%	100.0%	76.5%	30 or more	93.0%	64.0%	54.0%
26-29	0.0%	0.0%	11.8%	26-29	5.0%	8.0%	8.0%
21-25	0.0%	0.0%	11.7%	21-25	2.0%	21.0%	20.0%
20 or less	0.0%	0.0%	0.0%	20 or less	0.0%	7.0%	18.0%
	100.0%	100.0%	100.0%		100.0%	100.0%	100.0%

The Allan Gray study supports Bengen’s original study and finds that portfolios subject to an initial withdrawal rate of 4% would have lasted for at least 30 years, 93% of the time, and would have provided a retirement income for at least 26 years, 98% of the time. The Allan Gray study indicates that higher-than-inflation withdrawals would have had a significant negative impact on the longevity of portfolios even at an initial withdrawal rate of 4%.

Withdrawals assuming a 5% initial withdrawal would mean a 7% risk that the portfolio would last 20 years or less and a 21% chance that it would last between 21 and 25 years. This means that, to be safe, retirees should select an initial withdrawal rate of 4% in South Africa⁷. Another study in the USA by Cooley, Hubbard and Walz⁸ supported a higher withdrawal rate of up to 6% but this was based on returns from 1926 to 1995. **Page 2-47**

A study by Morningstar⁹ which focused on Australia but also analysed international returns for a number of countries, including South Africa, assumed a retirement period of 30 years, a portfolio of 50% equities and 50% bonds and went further than other studies by assuming an annual portfolio fee of 1% of assets.

The results for 99% and 95% success rates over 30-year retirement periods for the USA, the UK, Australia (AUS) and South Africa (RSA) are depicted in [Figure 2.10](#). For South Africa, this study found that to achieve a 99% success rate, the initial withdrawal rate would need to be about 3.2%.

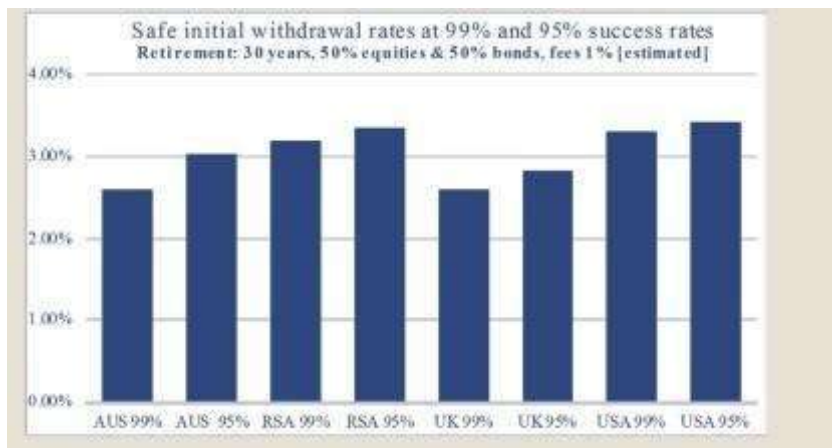


Figure 2.10 Safe initial withdrawal rates at 99% and 95% success rates

The Allan Gray study indicated that a 4% rule would lead to a 93% success rate for a 30-year retirement period but did not make an explicit adjustment for fees¹⁰ and assumed a greater proportion of equities at 55% of the portfolio. Generally, these studies support the principle of a low initial withdrawal rate to ensure that the real value of a pension annuity will be retained over a retirement period of 30 years. This is not only true for South Africa, but for other countries as well as set out in [Figure 2.10](#). It is important to note again that increasing the proportion of equities over lengthy retirement periods will tend to *reduce* the risk of outliving one's assets and running out of funds over the retirement period.

So where does this leave us? In our example for Lerato, we assumed a 10% portfolio return which seemed reasonable. However, this implied a risk that the portfolio would not last the full 30 years based on historical market returns and volatility as the withdrawal rate for year 1 was 6.65% of the opening value (R239400/R3602088). If we wish to be safe that Lerato's portfolio will outlast her and we set the initial annuity at 4%, then the first-year annuity will be R144000 rather than R239400, which is almost R100000 lower. Alternatively, in order to ensure that Lerato has an annuity of R239 400 in the first year, she would need to have saved very close to R6m (R239400/.04) in her retirement fund by retirement date. This indicates the starkness of the choices facing people going into retirement. Assume Lerato started with R3 602 088 and her initial withdrawal was R144000. So, what happened?

Scenario 1: Lerato had a happy but very frugal retirement and died at 95 years old. The markets provided the expected return of 10% and inflation was 5%. Lerato began with an initial 4% [Page 2-48](#) withdrawal and did not increase her withdrawals above inflation. This resulted in significant accumulation in her retirement waiting for a rainy day. When she passed away, she left a legacy of R25m to her heirs.

Scenario 2: Lerato had a happy but very frugal retirement and died at 95 years old. The markets experienced significant volatility and lower returns. Inflation was 5% per year and the portfolio return was only 6% per year over 30 years. Lerato began with an initial 4%

withdrawal and did not increase her withdrawals above inflation. When she passed away, she left a legacy of R218 000 to her heirs.

Limits on living annuities in terms of government regulations

In terms of government and SARS regulations, a living annuity withdrawal needs to be between 2.5% and 17.5% of the value of the retirement assets. We will ignore the 2.5% rule as this is normally not a constraint for most people. As the pension annuity increases due to inflation and the value of the investments and the annual return falls, then at some point the required inflation-adjusted annuity will come into conflict with the regulatory limit that the pension should not exceed 17.5% of the market value of the investments in any one year.

So, what is the effect of an underfunded pension plan in a regulated environment? What is the real difference between a beginning amount of R2.5m at the start of retirement and R3.6m if one wishes to earn a pension annuity of R228 000 in today's money over the next 30 years? What difference does a recommended withdrawal of 4% have in this case? The following reflects a situation whereby a South African retires with R2.5m and draws down R239 400 from year 1, which reflects an increase at an inflation rate of 5% per year. This reflects an initial withdrawal of 9.58% of the opening value of R2.5m. For year 1, the inflation adjusted annuity is R239 400 (R228 000 $\times$ 1.05). Whilst the retiree is able to legally withdraw 9.58% initially, the effect is that in real terms, after year 9, the actual annuity starts falling in order not to exceed the statutory limit of 17.5% of the value of the investments. This results in a steady reduction in the nominal and real value of the retiree's annuity over time, leading to lower living standards and even poverty. By year 20, the nominal value of the annuity should be R604 952 but the retiree will be receiving only R163 134. The limit of 17.5% ensures that retirees maintain an annuity for life but this falls dramatically over time. In year 15, the retiree is earning almost the same (R240 900) in nominal terms as at the end of year 1 but significantly less in real terms. In this case the golden years only lasted for 9 years. This situation is depicted in [Figure 2.11](#).

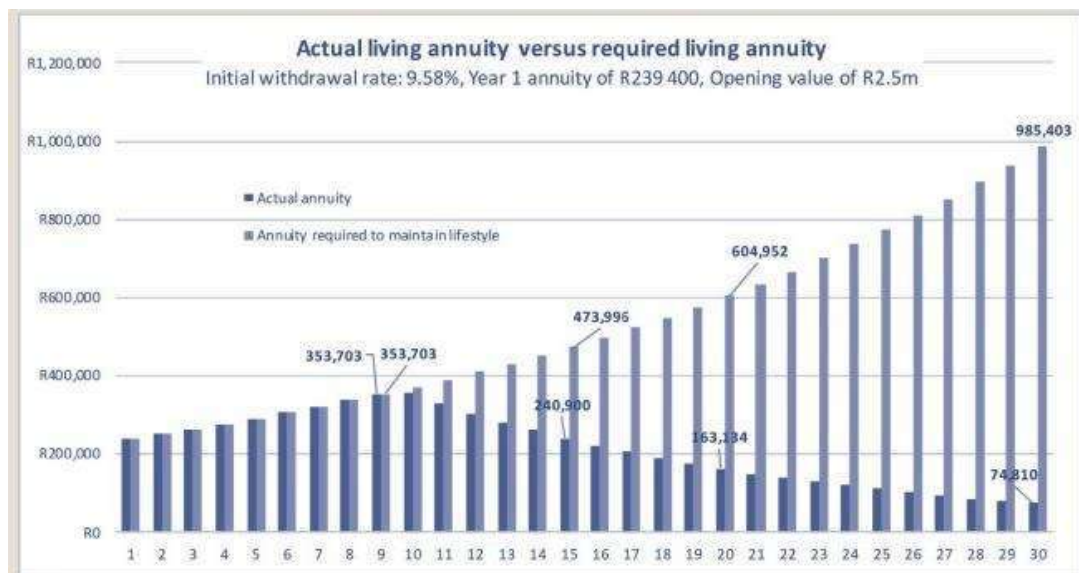


Figure 2.11 Actual annuity subject to 17.5% limit and the required annuity to maintain lifestyle per year

The nominal value of the pension by year 30 should be R985 403 per year in order to retain the purchasing power of R228 000 at the beginning of retirement. The effect of the 17.5% limit will mean that the pension will last at least 30 years and there will be ending capital of R395 426 but the pension annuity in year 30 will be R74 810, which is only 7.6% of the value that the pension annuity should be in order to maintain the same standard of living as today. **Page 2-49**

After 9 years, the pension annuity starts losing its real value and often pensioners are not ready for these adjustments. It is critical to act early and take drastic action if required. Otherwise, if you wait, it will be too late. It is important to plan from day one. To make the point more clearly, is to ask the question whether it would be desirable or possible for a retiree to live on R17 328 per year today (7.6% of R228 000)? That is the future one will face if one does not engage right away with the pension crisis which will probably occur in our lifetimes. You may consider that this situation will only affect you in the distant future, but it may affect your parents or family and it will therefore affect you.

[Table 2.12](#) sets out the impact of reducing the withdrawal rate. Assuming the portfolio offers an expected return of 10% per year (after fees) and inflation is 5%, then if the initial withdrawal rate is 4% to 6%, the annuity will last in **real terms** for at least 30 years but will have varying ending values. Whilst legislation permits a withdrawal of up to 17.5%, the reality is that investors and retirees should understand the negative impact that a high withdrawal rate will have on the future real value of their annuities.

Table 2.12 Withdrawal rates and the number of years a portfolio will last

Portfolio return: 10%, inflation 5%	4%	5%	6%	7%	8%	9%
Withdrawal rate						
No. of years annuity lasts in real terms	30	30	30	20	14	11
Success rate	100%	100%	100%	67%	47%	37%
Annuity for year 1	100,000	125,000	150,000	175,000	200,000	225,000
Ending portfolio value	17,368,585	10,804,856	4,241,127	1,173,531	638,706	452,491
Year that 17.5% limit applies	N/A	N/A	N/A	21	15	12

In [Table 2.12](#) we assumed that equity and bond markets will perform reasonably in the future based on current bond yields and expected equity returns. However, let's assume that the portfolio return is 8% rather than 10% per year. This now means that an initial withdrawal of 5% will offer a retiree a constant annuity in real terms for 26 years rather than 30 years. A withdrawal rate of 6% will offer an annuity in real terms that lasts for only 19 years rather than 30 years. A withdrawal rate of 7% will offer an annuity in real terms that lasts for only 14 years rather than 20 years. Only a withdrawal rate of 4% will ensure a constant annuity in real terms over 30 years if the portfolio return is 8%.

We suggested actions that Lerato could take in our example such as downsizing and moving into a smaller house and using the proceeds to invest to make up the difference in required annuity income. Otherwise, a person considering retirement should try and work for longer or take up part-time work or perhaps even rent out rooms and so on. Although these may be hard choices for some people, it is easier than what is to come if action is not taken early to deal with the pension crisis facing many people today.

We considered that it is important that the annuity increases by the inflation rate, but some costs such as healthcare have been increasing at a higher rate than inflation. Whilst we did not deal with asset allocation in any great depth, it is critical that retirees maintain at least a 50% exposure to equities over the retirement period and accept market volatility. Investing in bonds only, though less volatile, will have a negative impact over the long term on the value of the portfolio.

What would happen if Lerato starts with a much larger fund balance of R3.6m on retirement date, the portfolio provides a return of 10% per year and the inflation rate is 5% per year? In [Figure 2.12](#), we can see that, subject to the 17.5% limit, the pension annuity will continue to provide a real level of income until year 23. **Page 2-50**

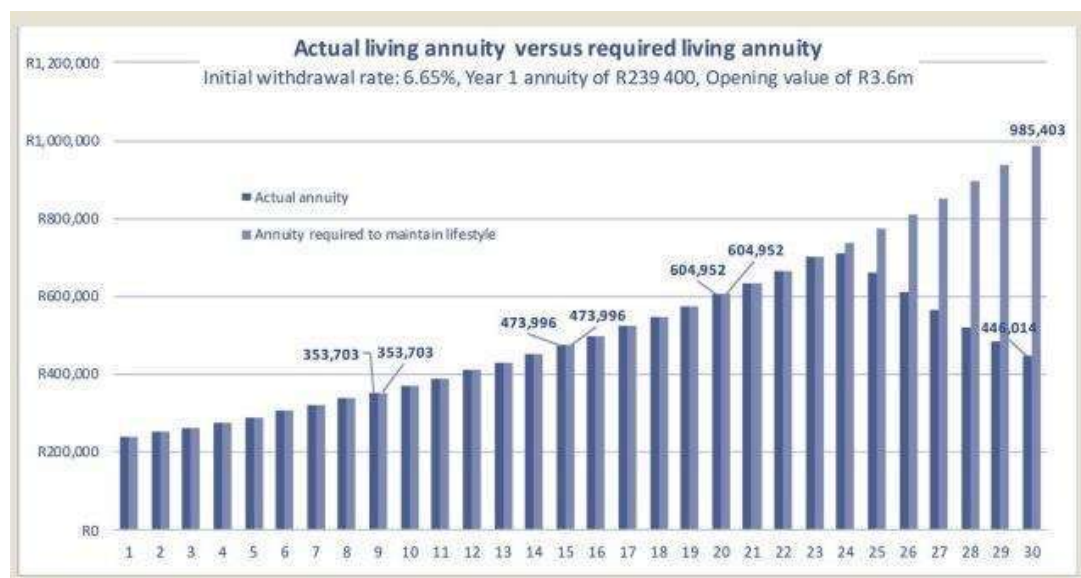


Figure 2.12 Actual living annuity subject to 17.5% limit and the required annuity to maintain lifestyle per year

It is only from year 24 that the pension annuity will fall in real terms as the annuity amount is constrained by the 17.5% limit on pension payments. Therefore, the annuity in year 30, due to this regulation, will be close to 50% of the required annuity value and there will be capital of about R2.36m at the end of year 30. The restriction of a 17.5% pay-out plays an important role to ensure that a retiree does retain retirement funds if the retiree lives longer than 30 years after retirement. However, this may come at the cost of a reduced annuity in real terms and our determination indicates that even a R3.6m funded living annuity means an annuity in

the 30th year that may be less than 50% of the required annuity if Lerato wishes to ensure that she maintains the same standard of living as when she retired. If Lerato's lifespan is 30 years after retirement, and the life expectancy tables indicate an expected remaining life of about 22 years for a woman at a retirement age of 65, then this means that there will be fund assets to leave to her heirs plus her home. Although the limit of 17.5% does protect to some extent against longevity risk, it may also impact on retirees such as Lerato and so it is important to evaluate a person's total financial position. There are ways to plan around this but this situation is not ideal and indicates the complexity of effective financial planning in retirement.

On 12 November 2017, Laura Du Preez in *Business Day* titled her article as: *SA retirees in living annuity 'crisis': Thousands of elderly have had pensions halved in six years*. The article reported that more than 20% of all retirees drawing a living annuity were receiving half the income, adjusted for inflation, as compared to six years previously. The Association for Savings and Investment South Africa (ASISA) responded by indicating that the average withdrawal rate was 6.62%.

A National Treasury technical discussion paper, *Enabling a better income in retirement*, published on 21 September 2012, states that charges on living annuities seem to be high at over 2%. The report states further that "a sustainable rate of drawdown for an individual aged 65 in good health may be no higher than 5% per year". The report indicates that there is a 2/3 chance that an annuitant's income will fall by 30% in real terms over an annuitant's life.

These reports indicate that the challenges facing South African retirees are real and it is important to keep to a withdrawal rate of 4% to 5%, in order to ensure that retirees do not end up financially distressed in their later retirement years. Living annuities have distinct advantages such as flexibility in relation to investments and the level of annuity income and there is transparency in relation to returns, investments and fees. However, there are real risks involved in relation to longevity, returns and inflation. Page 2-51

Is there a pension crisis on the horizon?

There are two main pension systems; a **defined contribution** plan and a **defined benefit** plan. Most companies in South Africa offer a defined contribution pension plan to their employees. In this case, there will be an accumulated sum on retirement which is determined by the contributions made to the fund by the employee and employer, plus the returns the fund is able to achieve on these contributions over the period of employment. This accumulated sum may then be used to acquire an annuity such as a living annuity or a guaranteed life annuity. The employee, and not the employer, is subject to investment risks during the accumulation phase and investment and longevity risks in the retirement phase, in the case of a living annuity.

In a defined benefit plan, the retirement fund will provide a pension to an employee based on a formula which considers the length of employment and the final salary in determining

the annuity to be payable to an employee for life. In most cases, the employee and employer contribute to the pension fund as a percentage of the employee's salary over the period of employment. The following would be an example of a defined benefit plan formula:

$$\text{Pension annual annuity} = \text{Final salary} \times 2\% \times \text{Number of years of service}$$

If the final salary was R480 000 per year and an employee has been working at the firm for 25 years, then the final pension annuity would be R240000 ($R480000 \times 2\% \times 25$). The trustees of the fund may decide to increase the annual pension by inflation but there may be no legal obligation to do so. In a defined benefit plan, the employee does not have to worry about investment performance or longevity risk. The employer takes on these risks.

The Government Employees Pension Fund (GEPF), which is a defined benefit fund, manages the pensions of all government employees in South Africa. It has more than 1.2 million active members and provides pensions currently to over 490000 pensioners and/or spouses. The GEPF has over R2.3 trillion assets under management and is the largest pension fund in Africa and one of the largest in the world. On retirement, a member will receive a once-off lump sum and an annuity. The annual annuity is based on 1/55 of the final salary multiplied by the number of years of pensionable service. The lump sum (gratuity) is 6.72% of the final salary multiplied by the number of years of pensionable service. The annuity will last over the life of the employee and thereafter 50% of the annuity will be paid to the employee's spouse over her or his life. The GEPF has been able to increase annuities in most years by inflation and is known to be a wellmanaged fund.

In most developed economies, the government will provide elderly people with an age pension. Public entities such as state governments in the USA will provide pensions to employees such as teachers and other public employees. In the USA and the UK, there are companies which have defined benefit plans, although companies have been moving across to defined contribution plans in order to reduce the investment and longevity risks implicit in defined benefit plans.

A World Economic Forum report in 2019 laid out the coming pension crisis facing developed economies. The pension crisis arises from promises made to citizens and employees that are not affordable nor sustainable. In terms of a Citi GPS report¹¹ in 2016, it was estimated that the value of unfunded or underfunded government pension liabilities for the 20 OECD countries was \$78 trillion, which Citi reported was almost double the national debt of \$44 trillion. USA and UK companies providing defined benefit plans have had to make increased contributions due to underfunded benefit plans.

The challenges facing governments and companies providing defined benefit plans relate to changing demographics and longevity as well as low interest rates. There is a growing increase in the 65+ aged population and in the number of people living longer, which results

in greater pension obligations. Low real interest rates and bond yields indicate that future real returns will be limited. **Page 2-52**

The move by central banks to raise interest rates in 2022 and 2023 have mostly reduced pension fund deficits as a higher discount rate will reduce the present value of a pension fund's liabilities. It has been estimated that in the USA, underfunding of state pension schemes amounts to about \$1.6 trillion.

How much companies need to contribute is a function of the expected rates of return on the investments held in retirement funds. The average assumed rate of return used in actuarial projections in 2022 for the top 100 USA companies in the S&P500 is 5.9%. The higher the investment rate, the less the company needs to contribute in the future. Is this assumption fair? If we assume that the investment portfolio will be 50% equities and 50% fixed-income securities and the current bond yield is offering 3.6% in the USA, then equities will need to offer an annual return of 8.2% over the longer term [$50\% \times 3.6\% + 50\% \times 8.2\% = 5.9\%$]. Is this reasonable? In the period 1900 to 2022, equities offered a real return of 6.4% and over the period 1973 to 2022, equities offered a return of 5.9% per year. If we add expected inflation in the USA of about 3%, it seems that an assumed return of 8.2% may be reasonable. Public pension funds on the other hand are estimating an annual portfolio return of about 7% and this may require equities to deliver a future compound return of 10.5% per year.

How does this affect South Africa? Firstly, most companies have shifted to defined contribution plans and therefore are not at risk. The risk has shifted over to the employees as we have set out in our example of a living annuity. Secondly, South Africa's bonds are offering higher long-term yields. If inflation adjustments are not obligatory, then this also provides protection to a firm if there are falls in returns. Also, the GEPP has been adept at managing the pension obligations of its members to date. However, if there are any significant bankruptcies in the USA then this may lead to disruption in financial markets which will not only affect the USA but will affect emerging markets like South Africa.

Although there may not be a pension crisis for the corporate sector in South Africa, our living annuity example indicates that for many pensioners, a combination of high withdrawals, poor investment decisions and living longer lives may lead to much lower living standards for retirees in later years. The golden years may lose their lustre due to the foreboding clouds of financial distress. It is important to act early – work longer, spend less, focus on your health, make your assets work for you such as your property, adhere to a low withdrawal rate in the early years of about 4% if possible, minimise fees unless there is outperformance after fees, and remain invested in equities even over periods of volatility. Acquire knowledge about investments, risk, investment managers and about how pensions and annuities work. It is important to know what is on the investment horizon in order to plan

effectively for retirement. We prepare for storms by trimming our sails in order to ensure a safe passage. Page 2-53

QUESTIONS

Question 2.1

How much will you accumulate if you are offered the following investments?

- R10 000 invested for 12 years at an interest rate of 8%, interest compounded annually.
- R12 000 invested for 6 years at an interest rate of 6%, interest compounded annually.

Question 2.2

How many years will it take for R1 000 to become R2 000 if an investment offers 6% per year, interest compounded annually?

Question 2.3

To what amount will regular payments of R1 500 per month at the start of each month accumulate by the end of two years if the payments are earning interest monthly at 6% per annum?

Question 2.4

You wish to travel to Europe in 5 years' time and you will need R30 000 at the time. If you currently have R15 000, what interest rate must you earn each year to reach R30 000?

Question 2.5

A bank is offering 6% per year, interest compounded monthly on a Special Savings account. If you deposit R8 000 today, how much will you accumulate by the end of 10 years? What is the annual effective rate?

Question 2.6

Calculate the future value of a R2 000 investment made now under each of the following interest rate and investment period alternatives. Assume annual compounding.

Investment	P	Q	R	S	T
Interest rate	8.0%	12.0%	10.0%	6.5%	13.5%
Number of years	4	6	5	10	3.5

Question 2.7

Use the investment alternatives in Question 2.6 but assume that semi-annual compounding is available on your initial investment. What will be the future value for each investment alternative? Also calculate the future values assuming quarterly compounding.

Question 2.8

A student borrowed R5 600 at the beginning of each year at 8% p.a. compound interest. How much is owed at the end of her third year?

Question 2.9

You are planning to take a holiday in Mauritius when your current savings of R10 000 reach R20 000.

- If you plan to take your holiday at the end of five years from now, what annual rate of interest will you have to earn on your savings account?
- If you are willing to wait seven years before taking your holiday, what annual rate of interest would be necessary on your savings account?
- If you can earn 7% per year compounded semi-annually on your savings account, how long will it take before you have adequate funds to take your holiday? Page 2-54

Question 2.10

If a lump sum of R8 000 left in the bank for four years has grown to R10 164 with interest compounding and paid at the end of each month, what annual rate of interest has the bank been paying?

Question 2.11

The manufactured price of an article was R2 880. Every time the article changed hands, the price was raised by 20%. If it was eventually sold for R5 972, how many times did it change hands?

Question 2.12

In terms of a bequest, a man can receive R2 250 immediately or R3 600 in 12 years' time. At what rate is compound interest earned, on an annual basis?

Question 2.13

At the birth of his daughter, a man invested R1 000 at 6% p.a. compounded annually. What amount would accrue to her if she left the money invested until she attained the age of

(a) 18 years, (b) 21 years, (c) 40 years, (d) 60 years?

Question 2.14

A special savings bond pays R30 in 10 years for each R10 invested today. At the time of his son's birth a man invests R5 000 in such bonds and at the end of 10 years reinvests the maturity value of the bonds at 12% p.a. compounded half-yearly. How much will be available to his son at age 18? Would it have been better for the father to have invested the original R5 000 immediately in the 12% investment?

Question 2.15

A bank agrees to lend you R10 000 today in return for your promise to pay the bank R18 380 in nine years' time. What rate of interest is the bank charging you?

Question 2.16

An individual has just put her life's savings into a bank account yielding 4% p.a., interest compounded semi-annually. Assuming she makes no withdrawals, how long must she wait until she has doubled her money?

Question 2.17

Mr A. Miller is a young financial director of a listed company. Although he enjoys his work, he wants to retire at the age of 35 (in 10 years' time) and go sailing in the Mediterranean around the Greek islands. He estimates that he will need to have R1.2m to buy the yacht and an additional R200 000 to pay for supplies and mooring costs. Mr Miller intends to make equal annual payments into a bank account on which he can earn 6% interest compounded annually.

- (a) What amount must Mr Miller pay annually to achieve his objective in 10 years' time? The first payment is to be made at the end of the first year.
- (b) Instead of making equal annual payments, Mr Miller wants to make one single sum payment today, investing it at 6% interest compounded annually. What should this single sum be?

Question 2.18

You borrow from Better Bank to buy a car, and make monthly repayments of R1 053.35 at 12% p.a. compounded monthly. Your loan is for a period of 4 years. How

Question 2.19

You want to travel to Brazil to visit friends when you graduate three years from now. The trip is expected to cost a total of R20 000 at that time. You have deposited R12 000 in an account paying 6% interest annually, maturing three years from now. You have inherited a further lump sum and plan to use it to finance the balance. If you are going to put this money in an investment earning 10% per year over the next three years, how much must you deposit now, so you can visit your friends at the end of three years?

Question 2.20

You can buy a music system for one cash payment today of R6 600 or ten quarterly payments of R750, starting at the end of three months. If you could invest the cash and earn 6% per annum, with quarterly compounding, would you prefer to pay in cash or make the quarterly payments?

Question 2.21

You have just received an inheritance of R329 760. You plan to put the entire amount in an account earning 8% compounded annually and to withdraw R40 000 at the end of each year. For how many years can you continue to make the withdrawals?

Question 2.22

You buy a studio apartment for R800 000 and put down a deposit of R200 000. Your repayments are R6 673.35 per month for the 15 years you intend to pay the apartment off in. What is the annual rate of interest on this loan?

Question 2.23

If your company borrows R2 000 000, repayable over ten years at 9% per annum with annual compounding and equal annual payments, how much of the loan will be outstanding after three years?

Question 2.24

A retirement plan guarantees to pay you or your estate a fixed annual amount for 20 years. At the time of retirement you will have R313 600 to your credit in the plan. The plan anticipates earning 10% interest annually over the period you receive benefits.

How much will your annual benefits be assuming the first payment occurs 1 year from your retirement date?

Question 2.25

WEC Ltd is planning to save R2 million per year for five years. The first deposit, which is presently being made, and all subsequent deposits, will earn interest at a 12% annual rate.

- (a) Calculate the future value for this annuity if interest is compounded semi-annually.
- (b) Calculate the future value for this annuity if interest is compounded quarterly.
- (c) How would your answer in part (a) have changed if the initial deposit was not made until the end of the first year?

Question 2.26

A student managed to acquire an interest-free loan of R80 000. After completing her studies and beginning to work, she saves R15 000 every year and invests the amount at 12.5% p.a. compound interest. What additional amount is required to fully repay the loan after four years of saving?

Question 2.27

In order to provide for R10 million to build a new warehouse in five years' time, a company plans to make equal payments at the end of each six months into a fund which earns 9% per year interest compounded semi-annually. After two years of payments, escalating costs lead the directors to increase the semi-annual deposit so that the fund will contain R12 million at the scheduled time of building. Find the increased semi-annual payment required for the remaining 3 years. Page 2-56

Question 2.28

A company has taken out a R480 000 loan. The loan is due at the end of six years, and the repayment amount is R779 648. What is the interest rate that the company is charged?

Question 2.29

A man wants to have R48 000 in eight years' time, so he decides to invest a certain sum half yearly at 14% p.a. interest compounded semi-annually. He invests the first sum at the end of the first six months and the same sum every half year thereafter. What would the difference in the instalment have been if he had paid the first instalment immediately?

Question 2.30

If a company wants to accumulate a total amount of R500 000 in 20 years' time, what annuity amount must it invest at the end of each year during the 20-year period? The applicable interest rate is 10% p.a.

Question 2.31

If you purchase an asset for R520.60 and it yields R100 per year for seven years, what is your rate of return on this investment?

Question 2.32

Assume a 13% RSA Government security (par value R100) with a redemption date of 15 July 2042, was quoting a yield to maturity of 8% on 16 July 2024. RSA Government Securities pay interest semi-annually. Interest is payable on 15 July and 15 January. Determine the market price of the 13% RSA security on 16 July 2024.

Question 2.33

A loan of R1.5 million was made to a farmer, bonded against fixed property at an interest rate of 8% p.a. compounded quarterly. The farmer agreed to pay off the loan in equal quarterly instalments over a period of 12 years. As a result of severe drought, at the end of five years, the farmer requested relief. It was agreed that the interest rate would be reduced to 4% p.a. By how much will the quarterly repayments of the farmer be reduced?

Question 2.34

Longlife Insurance Company agrees to buy the entire R100 million issue of 14% debentures of the Expand Manufacturing corporation, provided that the price will allow a yield of 8% per year compounded semi-annually. The debentures carry semi-annual interest and are redeemable at par in 20 years. How much cash does the corporation realise from the sale, if brokerage is 1% of the sale price?

Question 2.35

Calculate the annuity, payable in 12 annual instalments, which can be bought for R5 000 cash if the first instalment is paid immediately, and interest is compounded at 11% p.a.

Question 2.36

On his son's 12th birthday a father wants to invest a certain amount to enable him to withdraw R8 000 each quarter from his 17th to his 21st birthday. Calculate the sum he will have to invest if compounded interest is reckoned at 12% per annum, compounded quarterly.

Question 2.37

CC Company Ltd wants to establish a series of uniform deposits to be made on 1 January of 2x04, 2x05, 2x06 and 2x07 into a fund in order to make withdrawals of R2 million on 1 January 2x12, 2x13 and 2x14. What must the size of these deposits be to enable CC to make withdrawals if the fund will earn interest at the rate of 12% p.a. interest compounded annually?

Question 2.38

What is the present value today of a perpetuity that pays R400 per year if the first payment does not begin until four years hence and if 10% is the relevant discount rate? Page 2-57

Question 2.39

A machine is bought with a loan which must be repaid in 20 equal half-yearly instalments of R6 500. The first instalment is payable four years after the loan has been negotiated. What is the price of the machine if compound interest is added every half year at 14% p.a. on the outstanding amount?

Question 2.40

A firm's earnings per share grew from R5.78 to R12.48. During the same ten-year period, its sales grew from R60 million to R156 million. Calculate the difference in growth rates between EPS and sales.

Question 2.41

A non-redeemable preference share pays a R15 dividend each quarter. What is the maximum price an investor should pay for the share if a yield of 8% per year is required?

Question 2.42

Normal carpeting costs a company R280 per running metre and lasts five years. Is it more economical to purchase a special fibred carpet at R420 per running metre which will last eight years if the money is worth 12% p.a.?

Question 2.43

What is the value of an investment at the end of 20x9 if you intend to contribute R10 000 at the beginning of each year from 20x5 to 20x9? You can earn an interest rate of 6% p.a. (interest compounded monthly).

Question 2.44

You want to retire in 30 years and will require a monthly income of R20 000 for 10 years after retirement date. If the return you can obtain is 12% p.a. (interest compounded monthly), how much must you contribute every month for the next 30 years in order to obtain an annuity of R20 000 per month for 10 years after retirement date?

Question 2.45

You bought a small apartment for R950 000 on 1 February 20x4 (registration date) and you obtained an 80% mortgage bond on that date. The bond interest rate since 20x4 has stayed constant at 7.2% per annum, interest compounded monthly. The rate of 7.2% is the actual return that the bank is earning. Bond repayments are payable in advance on the first day of the month. The mortgage bond had a term of 20 years from 1 February 20x4. What is the capital balance outstanding on your bond on 1 April, 20x13 (before April's bond repayment)?

Question 2.46

You have started a small manufacturing company. You have bought a machine in terms of a suspensive sales agreement whereby you are required to make equal monthly instalments from today, 1 April 2x08, to 1 March 2x13. The cash price of the machine is R220 000. Finance charges are linked to the prime overdraft rate. The bank will charge you a premium of 1% above the prime rate which is currently 11%.

Required:

- (a) Determine the equal monthly instalment amount required to purchase the machine over five years.
- (b) If the prime rate increases to 14% today, before your first instalment payment, what will be the increase in your monthly instalment amount?
- (c) The bank offers to give you a three-month "holiday" so that you acquire the machine today but the monthly repayments begin on 1 July 2x08. The last payment is still on 1 March 2x13. What will your monthly instalment be if the prime rate is currently 11%?

Question 2.47

SX Company has issued 3 353 793 cumulative redeemable preference shares. The following is an extract from SX's annual financial statements for the year ending 30 June 2x04.

The cumulative redeemable preference shares carry a dividend of 12.5% per annum on the issue price of R1.00 per share and will be redeemed at a price of R1.00 per share in five equal annual instalments, payable on 31 December each year, commencing on 31 December 2x05.

Assume the preference dividend is payable annually on 31 December. Investors require a return of 9% p.a. (interest compounded annually) on similar preference shares.

Required:

Determine the price you would be prepared to pay for 100 SX Ltd cumulative preference shares. Assume the current date is 31 December 2x04 (ex. dividend). Your share will be redeemed in five equal annual instalments.

Question 2.48

You have purchased a pre-owned motor vehicle in terms of a suspensive sales agreement. You are required to pay monthly instalments from today, 1 June, 2x01 to 1 May 2x06. The cash price of the car is R128 496.00. Finance charges are linked to the prime overdraft rate; the rate is prime plus a premium of 1% per annum. The prime rate is currently 8% per annum, interest compounded monthly.

Required:

- (a) Determine the equal monthly instalment amount required to purchase the motor vehicle over five years.
- (b) If in a year's time, on 31 May 2x02, the prime rate increases to 11%, what will be the new monthly instalment amount from that date?

Question 2.49

Mirton Ltd has non-cumulative, non-redeemable preference shares in issue. The issue price was R1.00 each and the coupon preference dividend rate is 12% per annum, payable once a year in arrears. The company has not paid out a dividend in recent years but expects to recommence dividend payments in two years' time from today.

What is the value of each preference share if similar preference shares are quoting yields of 9% per annum?

Question 2.50

Mr Smart has just turned 50 years old and wishes to plan for his retirement at the age of 60 in 10 years' time. He is considering either investing R90 000 at the end of each year over this 10- year period in a mutual fund or investing R150 000 per annum in an all-equity retirement fund. Mr Smart is on a marginal tax rate of 40% and his annual contribution to the retirement fund will be fully tax deductible. You may assume that any tax benefit will be realised at the same time as the contribution during the initial 10-year period. His annual contribution to the mutual fund is not tax deductible. In 10 years' time, the total accumulated amount in the mutual fund may be withdrawn and is subject to tax at a rate of 10% of the difference between cost and the selling price. If Mr Smart selects the retirement fund option then the accumulated amount must be used to acquire a 10-year annuity which is taxable.

When Mr Smart retires, he expects his marginal tax rate to decrease to 30%. The financial institution expects both the retirement fund and the mutual fund to grow at an annual compounded rate of 9% over the initial period. The financial institution will be able to acquire a monthly annuity on Mr Smart's retirement, which will result in a return of 9% per annum, interest compounded monthly. This reflects Mr Smart's required return on funds invested during his retirement period. You may assume that tax is payable once a year, at the end of each year. You may assume that differences in transaction costs for the two options are immaterial. Mr Smart is in a position that he will not pay tax on any mutual fund income. The mutual fund income is immediately reinvested, and forms part of the 9% annual return. **Page 2-59**

Required:

Determine whether Mr Smart should select the retirement fund option or the mutual fund option based on the value relative to cost of each alternative on retirement date if he wishes to maximise his after tax wealth.

Question 2.51

You are considering investing in a fund whereby you will receive one third of the accumulated sum as a single sum payment in 30 years' time on retirement date, and a monthly income of R24 000 for 10 years subsequent to retirement date. You will

undertake the policy through one of the major financial institutions which can earn a return of 6% per annum, interest compounded monthly.

Required:

Determine the monthly contribution that you are required to make over the next 30 years to obtain the single sum payment on retirement date and a monthly income of R24 000 for 10 years.

Question 2.52

Two years ago, A Ltd needed to accumulate a total of R600 000 by the end of four years to acquire new imported plant and machinery from an Italian supplier. To do so, A Ltd makes equal semi-annual deposits into a fund which earns 8% per annum, interest compounded semi-annually. To date, the company has made four equal semi-annual deposits into the fund. The rand has recently declined against the Euro, and to acquire the plant and machinery, A Ltd now requires to accumulate a total of R1 000 000 in two years' time. However, A Ltd can from now on earn 10% per annum, interest compounded semi-annually. All monies accumulated to date will from now on earn a return of 10% per annum, interest compounded semi-annually. What is the increased equal semi-annual deposit that A Ltd will need to make so that the fund will contain R1 000 000 in two years' time?

Question 2.53

A 6-year bond, an RSA Government Security with a nominal value of R100 000 and coupon rate of 9% is issued today, when the YTM required by investors is 9%.

Required:

- (a) Calculate the present value of the stream of coupon payments, excluding the nominal face value payable on maturity.
- (b) Calculate the present value of the nominal face value of the bond.
- (c) Sum the two values obtained in (a) and (b) above, and briefly explain the reasons for the amount that you have calculated.
- (d) Without further calculation, estimate what the value of the bond will be in three years' time, if the YTM remains unchanged, and explain your choice of estimate.

Question 2.54

A 20-year RSA Government Security with a nominal value of R10 000 and coupon rate of 6% was issued 15 years ago, when the YTM required by investors was 6%.

Required:

- (a) Calculate the value of the bond 12 years ago, when the YTM was 6%.
- (b) Calculate the value of the bond today, if the required YTM is 12%.
- (c) If the YTM changed to 10% today, as a result of a decrease in interest rates in the economy, calculate the value of the bond today using a YTM of 10%.

Question 2.55

As a bond portfolio manager, you purchase a 10% RSA Government Security with a nominal value of R10 million. It has 7 years to maturity. The yield to maturity (YTM) at the date of purchase is 12%, which is immediately after the coupon has been paid. You hold the bond for exactly 1 year, receiving the next coupon, and then sell the bond. The YTM at the date of sale has fallen to 9%. Page 2-60

Required:

- (a) Calculate the amount received from the coupon payment.
- (b) Calculate the price that you paid for the bond.
- (c) Calculate the price for which you sold the bond.
- (d) Explain the reason for the profit or loss which was made over the one year holding period, with reference to the principles underlying the time value of money.

¹ We are grateful to Colin Smith for making this point.

² In terms of market convention, we quote rates on an annualised basis (interest rate per period $\times$ number of periods in a year). This is not the effective annual rate, which is $(1.04)^2 = 8.16\%$.

³ Almost all employees in the private sector will belong to a defined contribution retirement plan but government employees will belong to a defined benefit plan. We explain the difference between these plans later in this appendix. For now, we will only refer to members of defined contribution retirement plans.

⁴ If we consider a long-term bond yield of 8%, a market risk premium of 4% to 5% which means that future equity returns should be about 12% per year, and an asset allocation of 50% equities and 50% bonds, then an annual return for the portfolio of about 10% seems reasonable. We will assume that this return is after fees. Inflation should be around 5% in the longer term.

⁵ William P. Bengen. Determining withdrawal rates using historical data, *Journal of Financial Planning*, 171180 [p173], October 1994.

⁶ Michael Summerton, “How to ensure your retirement income goes the distance”, Allan Gray Quarterly Commentary, 2, 2014, 13–16.

⁷ Financial planners may alternatively refer to the Rule of 300 which is the monthly income required to cover monthly living costs multiplied by 300. The monthly Rule of 300 is related to

the annual 4% rule $[1/(0.04/12)]$.

⁸ Cooley, PL, Hubbard, CM and Walz, DT. “Retirement Savings: Choosing a withdrawal rate that is sustainable”, *AAIL Journal*, Feb. 1998.

⁹ Blanchett, D, Serhan, A, and Gee, P. “Safe withdrawal rates for Australian retirees”, Web: 2 February 2016, <https://www.morningstar.com.au/smsf/article/withdrawal-rates/7529?q=printme>

¹⁰ It is important in our view to consider fees which may be in the order of around 1.0% to 2.5% for most actively managed funds in South Africa. However, we need to consider this in line with the performance (and risk) of each fund relative to the market benchmark. Therefore, if a fund manager can achieve a return that matches the market benchmark **after** fees (VAT inclusive), then we think that it is more correct to assume a higher withdrawal rate than indicated by Morningstar and therefore the 4% rule is probably more relevant. If the performance of the fund is equal to the market index before fees, then it is relevant to include an estimate for fees. The bottom line is to use an initial 4% withdrawal rate or lower and to increase the equity component of the portfolio to ensure a safe financial passage over one’s retirement.

¹¹ The Coming Pensions Crisis: Recommendations for keeping the global pensions afloat, Citi GPS: Global Perspectives & Solutions.